

An XFEM-strain gradient damage model for efficient modeling of materials with reinforcement particles

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ABSTRACT

A coupling of extended finite element method (XFEM) and strain gradient enhanced damage method is proposed to model damage initiation and growth in the materials with reinforcement particles. The proposed model inherits the advantages of both methods i.e., simple definition of reinforcement particles from XFEM and numerical stability during damage evolution from the strain gradient damage model. The capability of the proposed XFEM-strain gradient damage model has been demonstrated by simulating mode-I, mode-II, and the mix-mode failure mechanism in the SENB, SEN, and L-shaped specimen with reinforcement particles respectively. The numerical simulations are performed by considering hard and soft reinforcement particles in a matrix. Mesh-sensitivity and particle-sensitivity analysis are performed to quantify effect of various parameters like mesh size, particle size and volume fraction on behavior of SENB specimen. The proposed XFEM model is at least 10% efficient (at higher mesh densities) than the standard FEM model for solving the SENB specimen with reinforcement particles. In the particle sensitivity analysis, proposed XFEM model used only one mesh density (unlike FEM model that needs different conformal meshes) to simulate set of 400 realizations with random spatial distribution of reinforcement particles. The analysis shows that the computational demand for the creation of conformal mesh is alleviated with the use of XFEM for defining reinforcement particles. The mesh-sensitivity analysis shows that difference in the solution obtained using proposed XFEM model and standard FEM model is negligible (less than 1% at higher mesh densities). Therefore, the proposed XFEM-strain gradient damage model can substitute FEM-strain gradient damage model with above mentioned efficiencies.

1. Introduction

The prediction of structural and damage response in heterogeneous materials is of paramount importance due to their wide spectrum of applications in the structural, mechanical, robotics and biomedical fields [24]. However, due to their inhomogeneous structure, it is very difficult to accurately estimate their response using conventional theories and models. Therefore, numerous models have been proposed in the recent past to estimate the structural and damage response in heterogeneous materials [23]. Among these

Abbreviations: BC, Boundary condition; DOFs, Degrees of freedom; FEM, Finite element method; SENB, Single edge notch beam; SEN, Single edge notched; XFEM, Extended finite element method.

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Nomenclature

Greek Letters

Γ	External boundary of a problem domain
Γ_t	Boundary of a problem domain with prescribed traction
Γ_d	Boundary of weak discontinuity in a problem domain
Γ_u	Boundary of a problem domain with prescribed displacement
Ω	Problem domain
ϵ	Strain tensor
ϵ_{eq}	Equivalent strain
$\bar{\epsilon}_{eq}$	Non-local equivalent strain
σ	Cauchy's stress tensor
ζ	Relative position vector used to define the local averaging domain
ψ	Ridge enrichment function
ψ_a	Area under the load-displacement curve
κ	Scalar strain history parameter
κ_0	Threshold strain parameter
$\epsilon_{xx}, \epsilon_{yy}$	Uniaxial strain in the x-direction and y-direction respectively, mm/mm
γ_{xy}	Shear strain in xy-plane, mm/mm
ξ, η	Parametric coordinates representing abscissa and ordinate respectively
$\bar{\xi}, \bar{\eta}$	Sub-parametric coordinates representing abscissa and ordinate respectively

Conventions

$\mathbf{C}$	Hookean stiffness tensor
D	Damage variable
F_{max}	Peak load
I_1	First invariant of the strain tensor
$\mathbf{t}$	Prescribed traction of the boundary

Material constants

c	Gradient parameter characterizing interaction domain
E	Elastic modulus
k_r	Ratio of compressive strength to the tensile strength of the material
l	Averaging length parameter
α, β	Material parameters for exponential softening law
λ_1, λ_2	Lamé constants
μ	Shear modulus
ν	Poisson's ratio

Operator

∇	Gradient operator
δ	Variation operator
$\sum$	Summation operator

Global matrices

$\mathbf{B}$	Matrix containing derivatives of quadratic shape functions
$\bar{\mathbf{B}}$	Matrix containing derivatives of linear shape function
$\mathbf{f}$	Force vector
$\mathbf{K}_{aa}$	Stiffness matrix based on standard degrees of freedom
$\mathbf{K}_{\epsilon\epsilon}$	Local equivalent strain-based stiffness matrix
$\mathbf{K}_{a\epsilon}$	Coupled stiffness matrix based on standard and local equivalent degrees of freedom
$\mathbf{K}_{bb}$	Stiffness matrix based on enriched degrees of freedom
$\mathbf{K}_{ua}$	Coupled stiffness matrix based on standard and enriched degrees of freedom
$\bar{\mathbf{N}}$	Shape function matrix of four-node linear quadrilateral element
$\mathbf{N}$	Shape function matrix of eight-node quadratic quadrilateral element

Global vectors

$\tilde{\mathbf{u}}_u$	XFEM nodal displacement vector
$\tilde{\mathbf{u}}_a$	Standard nodal displacement vector
$\tilde{\mathbf{u}}_b$	Enriched nodal displacement vector
$\tilde{\epsilon}_{eq}$	Non-local equivalent strain nodal vector

Subscripts and Superscripts

A	Superscript index used to represent analytical solution
a	Subscript representing quantities associated with standard nodes of finite element mesh
b	Subscript representing quantities associated with enriched nodes of finite element mesh
e	Superscript and subscript representing element number in finite element mesh
ext	Subscript index used to represent external vector
i	Superscript index used to represent i^{th} iteration
int	Subscript index used to represent internal vector
max	Subscript index used to represent the maximum quantity
u	Subscript representing quantities associated with standard and enriched nodes
ε	Subscript index representing quantities associated with non-local equivalent strain

Abbreviations

BC	Boundary condition
DOFs	Degrees of freedom
FEM	Finite element method
SENB	Single edge notch beam
SEN	Single edge notched
XFEM	Extended finite element method

methods, the focus of the present work is on the continuous-discontinuous methods for modeling material with reinforcement particles.

The modelling of reinforcement particle is relatively more complex because a conformal mesh is generally needed for its accurate representation. However, building a conformal mesh is time consuming and may not be always viable. Particularly, when the number of reinforcement particles is very large, it is very difficult and computationally expensive to generate a conformal mesh for each particle. Therefore, recently several methodologies have been developed to use the extended finite element method (XFEM) for efficient modelling of heterogeneities [18,5,6,22,32].

Despite being very efficient in modelling discontinuities (heterogeneities and crack), the XFEM becomes very difficult to handle multiple level sets for heterogeneities and crack simultaneously [1]. Therefore, to overcome this difficulty, several models have been developed that combine the continuous (gradient damage method/phase field method) and discontinuous methods (XFEM). In this regard, there have been several publications that focus on combining the continuous and discontinuous methods through the combination of the extended finite element method (XFEM) and gradient damage method (GDM) [34]. These methods can be classified into two categories, i.e., with and without cohesive zone transition. The cohesive zone transition in these methods uses a traction separation law to make the transition from GDM to XFEM upon complete failure. The cohesive zone transition-based methods can be found in the works of Comi et al. [8], Cazes et al. [9], Cazes et al. [10], Seabra et al. [37], Roth et al. [33], Wang and Waisman [40], Jin and Arson [17] and Tamayo-Mas et al. [38]. The other classification is without the cohesive zone transition, which includes the works of Mazars and Pijaudier-Cabot [20], Patzák and Jirásek [25] and Simone et al. [36]. A more recent work by Geelen et al. [12] and Patil et al. [26] uses the phase field method as the continuous approach instead of GDM.

It must be noted that in all the above-mentioned methods except [26], a complete transition from a continuous to a discontinuous method is carried out. It is also evident that this transition is not very accurate [36] and often needs some additional complexities to be reasonably accurate [33]. Therefore, in the present work, it is proposed to use the XFEM for modelling reinforcement particles and GDM for crack initiation/propagation. In other words, the reinforcement particles in the model are defined using XFEM (enrichments and level sets) while the crack/damage will be defined by the gradient damage model whose formulation is based on the non-local damage theory [42]. The proposed approach is different from the other mesh-free discontinuous methods like the cracking particle method [29–31] in the sense that it does not use any enrichments for the definition of crack. The proposed model will have the following advantages over the conventional methods for modeling material with reinforcement particles,

- The computationally expensive task of tracking crack discontinuities (for an XFEM only model) will be alleviated due to strain-gradient damage model.
- The computationally demanding creation of conformal mesh is alleviated when the XFEM is used for the definition of reinforcement particles.
- The complexities associated with the cohesive zone transition in the existing continuous-discontinuous methods are alleviated as there is no transition in the proposed model.

The paper is organized as follows: In Section 2 details regarding the numerical formulation of the proposed model have been presented. Various crucial sub-sections such as variational formulation, linearization of weak governing equations, and numerical integration are also presented in Section 2. The numerical results and discussions are elaborated under Section 3. Section 4 lists the conclusions along with limitations and future scope of the present study. Additionally, the expressions of von-Mises equivalent strain and its derivatives are presented in Appendix A. Details of non-local equivalent strain are presented in Appendix B. Lastly, Appendix C contains the expression of quadratic and linear shape functions used for approximation of displacement and non-local equivalent strain

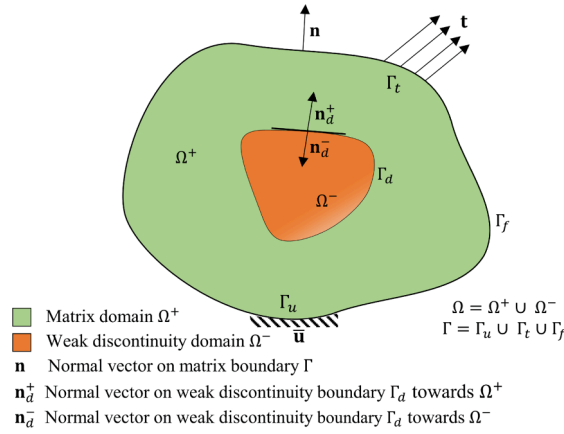


Fig. 1. A Schematic representation of the domain Ω where, Ω^+ is the matrix domain and Ω^- is the heterogeneity domain such that $\Omega = \Omega^+ \cup \Omega^-$. The domain Ω is bounded by Γ where, displacement $\bar{\mathbf{u}}$ is prescribed on Γ_u , tractions $\mathbf{t}$ are prescribed on Γ_t and free.

respectively.

2. Coupled XFEM-Gradient damage model

In this section, the numerical formulation of the weakly coupled XFEM-Gradient damage model has been presented.

2.1. Governing equation

The strong form of governing equation in a two-dimensional body Ω (Fig. 1) with an internal weak discontinuous boundary Γ_d without body forces can be written as,

$$\nabla \cdot \boldsymbol{\sigma} = 0, \quad (1)$$

where ∇ is the gradient operator, $\boldsymbol{\sigma}$ is the Cauchy stress tensor. The boundary conditions are defined as follows,

$$\mathbf{n} \cdot \boldsymbol{\sigma} = \mathbf{t} \text{ on } \Gamma_t \subset \Gamma, \quad (2)$$

$$\mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_u \subset \Gamma, \quad (3)$$

$$\mathbf{u}^+ \mathbf{t}_d^+ = \mathbf{u}^- \mathbf{t}_d^- \text{ on } \Gamma_d, \quad (4)$$

where $\mathbf{t}$ is the prescribed traction on the boundary Γ_t . $\bar{\mathbf{u}}$ is the prescribed displacement on the boundary Γ_u . $\mathbf{n}$ is the unit outward normal vector to the external boundary Γ such that $\Gamma_u \cup \Gamma_t \cup \Gamma_f = \Gamma$ and $\Gamma_u \cap \Gamma_t = \emptyset$. $\mathbf{t}_d^+$ and $\mathbf{t}_d^-$ is the traction on boundary of weak discontinuity Γ_d toward $\mathbf{n}_d^+$ and $\mathbf{n}_d^-$ respectively.

The expression for stress-strain relation for isotropic damage is given as,

$$\boldsymbol{\sigma} = \mathbf{C}(1 - D)\boldsymbol{\varepsilon}, \quad (5)$$

where $\mathbf{C}$ is the Hookean stiffness tensor and $\boldsymbol{\varepsilon}$ is the strain tensor. The expression for Hookean stiffness tensor ($\mathbf{C}$) for plane strain problem is given by,

$$\mathbf{C} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & 0.5-\nu \end{bmatrix}, \quad (6)$$

where E is the modulus of elasticity and, ν is Poisson's ratio.

The expression of scalar damage variable (D) based on exponential softening law is given as,

$$D = \begin{cases} 0 & \text{if } \kappa \leq \kappa_o, \\ 1 - \frac{\kappa_o}{\kappa} [1 - \alpha + \alpha \exp(-\beta(\kappa - \kappa_o))] & \text{otherwise,} \end{cases} \quad (7)$$

where κ_o is the threshold strain beyond which damage is initiated. κ is the scalar strain history parameter of a material point. The value of κ is equal to the maximum value of local equivalent strain (ε_{eq}) at material point throughout deformation history. α and β are

exponential softening parameters. β decides the steepness in the damage evolution i.e. rate of change in damage w.r.t. change in deformation (strain).

Modified Helmholtz equation for non-local integral averaging of the equivalent strain ε_{eq} is written as,

$$\bar{\varepsilon}_{eq} - c\nabla^2 \bar{\varepsilon}_{eq} = \varepsilon_{eq}, \quad (8)$$

where c is the gradient parameter and $\bar{\varepsilon}_{eq}$ is non-local equivalent strain. The size of non-local averaging region is equal to c^2 . The details of von-Mises equivalent strain (ε_{eq}) and non-local equivalent strain ($\bar{\varepsilon}_{eq}$) are mentioned in [Appendix A](#) and [Appendix B](#) respectively. The boundary condition for solving the modified Helmholtz equation is,

$$\nabla \bar{\varepsilon}_{eq} \cdot \mathbf{n} = 0. \quad (9)$$

In the gradient damage formulation, it ensured that the tangent stiffness tensor also known as acoustic tensor [\[16\]](#) is non-singular (has positive eigenvalues) during strain localization. The positive definiteness of tangent stiffness tensor is ensured by the non-local regularization that acts as a localization limiter and preserves the ill-posedness of the BVP [\[15\]](#).

2.2. Variational formulation

The governing equation is transformed by multiplying with displacement-based weight function ($\mathbf{w}_u$) and integrating over the domain Ω , we get,

$$\int_{\Omega} \mathbf{w}_u \cdot (\nabla \cdot \boldsymbol{\sigma}) d\Omega = 0, \quad (10)$$

Applying Gauss divergence theorem and substituting expression of Cauchy's stress tensor i.e. $\mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n}$, we get the final form of Eq. [\(10\)](#) as,

$$\int_{\Omega} \nabla^s \mathbf{w}_u : \boldsymbol{\sigma} d\Omega = \int_{\Gamma} \mathbf{w}_u \cdot \mathbf{t} d\Gamma, \quad (11)$$

Similarly, the equivalent strain boundary condition in Eq. [\(9\)](#) is transformed by multiplying with strain-based weight function ($\mathbf{w}_\varepsilon$) and integrating over the boundary Γ we get,

$$\int_{\Gamma} \mathbf{w}_\varepsilon \cdot \nabla \bar{\varepsilon}_{eq} \cdot \mathbf{n} d\Gamma = 0, \quad (12)$$

Applying Gauss divergence theorem and substituting in modified Helmholtz equation, we get the final form of Eq. [\(12\)](#) as,

$$\int_{\Omega} \mathbf{w}_\varepsilon \cdot \bar{\varepsilon}_{eq} d\Omega + \int_{\Omega} c \nabla \mathbf{w}_\varepsilon \cdot \nabla \bar{\varepsilon}_{eq} d\Omega = \int_{\Omega} \mathbf{w}_\varepsilon \cdot \varepsilon_{eq} d\Omega. \quad (13)$$

The variation of stress tensor $\delta \boldsymbol{\sigma}$ using Eq. [\(5\)](#) can be written as,

$$\delta \boldsymbol{\sigma} = (1 - D) \mathbf{C} \delta \varepsilon - \mathbf{C} \varepsilon \delta D. \quad (14)$$

The variation in damage variable (δD) based on Eq. [\(7\)](#) can be written as,

$$\delta D = \left(\frac{\partial D}{\partial \kappa} \right) \delta \bar{\varepsilon}_{eq}, \quad (15)$$

where the expression of $\frac{\partial D}{\partial \kappa}$ is given as,

$$\frac{\partial D}{\partial \kappa} = \begin{cases} \frac{\kappa_o}{\kappa^2} (1 - \alpha + \alpha e^{-\beta(\kappa - \kappa_o)}) + \frac{\alpha \beta \kappa_o}{\kappa} e^{-\beta(\kappa - \kappa_o)} & \text{if } \kappa \geq \kappa_o, \\ 0 & \text{if } \kappa \leq \kappa_o, \end{cases} \quad (16)$$

Substituting Eq. [\(15\)](#) into Eq. [\(14\)](#), the variation of stress tensor can be written as,

$$\delta \boldsymbol{\sigma} = (1 - D) \mathbf{C} \delta \varepsilon - \mathbf{C} \varepsilon \left(\frac{\partial D}{\partial \kappa} \right) \delta \bar{\varepsilon}_{eq}. \quad (17)$$

2.3. Discretization and linearization weak governing equation

In this section, the linearized form of the governing equation for a body with a weak discontinuity is presented. In the following, the subscript u and ε is used to represent displacement and non-local equivalent strain field.

2.3.1. Discretized field variables

The displacement field is discretized using a quadratic shape function of an 8-noded quadrilateral element. The discretized displacement field is expressed as,

$$\mathbf{w}_u = \mathbf{N}_a \tilde{\mathbf{u}}_u = \mathbf{N}_a \tilde{\mathbf{u}}_a + \mathbf{N}_b \tilde{\mathbf{u}}_b, \quad (18a)$$

$$\nabla^s \mathbf{w}_u = \mathbf{B}_a \tilde{\mathbf{u}}_u = \mathbf{B}_a \tilde{\mathbf{u}}_a + \mathbf{B}_b \tilde{\mathbf{u}}_b, \quad (18b)$$

where $\tilde{\mathbf{u}}_a$ and $\tilde{\mathbf{u}}_b$ is the nodal displacement vector representing standard and enriched degrees of freedom respectively. $\mathbf{N}_a$ is the matrix containing quadratic shape function of nodes in an element with standard degrees of freedom (refer to [Appendix C](#)). $\mathbf{N}_b = \psi \mathbf{N}_a$ is the matrix containing quadratic shape function of nodes in an element with enriched degrees of freedom. ψ is the ridge enrichment function [\[43\]](#). $\mathbf{B}_a$ and $\mathbf{B}_b$ is the matrix (for details of XFEM B-matrix refer [\[3\]](#)) containing derivatives of quadratic shape functions, $\mathbf{N}_a$ and $\mathbf{N}_b$ respectively.

The weight function ($\mathbf{w}_\varepsilon$) for non-local equivalent strain is discretized with linear interpolation functions to meet the consistency requirements for the quadratic rate of convergence. The weight function ($\mathbf{w}_\varepsilon$) is expressed as,

$$\mathbf{w}_\varepsilon = \bar{\boldsymbol{\varepsilon}}_{eq} = \bar{\mathbf{N}} \bar{\boldsymbol{\varepsilon}}_{eq}, \quad (19a)$$

$$\nabla \mathbf{w}_\varepsilon = \nabla \bar{\boldsymbol{\varepsilon}}_{eq} = \bar{\mathbf{B}} \bar{\boldsymbol{\varepsilon}}_{eq}, \quad (19b)$$

where $\bar{\mathbf{N}}$ is the matrix containing shape functions of the 4-node quadrilateral element (refer to [Appendix C](#)). $\bar{\mathbf{B}}$ is the matrix containing spatial derivatives of the linear shape function.

2.3.2. Discretized weak equations

The discretized form of the weak governing equation can be written as follows,

$$\int_{\Omega} \mathbf{B}_u^T \boldsymbol{\sigma} d\Omega = \int_{\Gamma} \mathbf{N}_u^T \mathbf{t} d\Gamma, \quad (20)$$

The equivalent nodal force vector for admissible variation of $\tilde{\mathbf{u}}_u$ can be written as,

$$\mathbf{f}_u^{int} = \int_{\Omega} \mathbf{B}_u^T \boldsymbol{\sigma} d\Omega, \quad \mathbf{f}_u^{ext} = \int_{\Gamma} \mathbf{N}_u^T \mathbf{t} d\Gamma. \quad (21)$$

Substituting Eqs. (19a) and (19b) to Eq. (13), the final expression of a discretized weak form of the Helmholtz equation is written as,

$$\left(\int_{\Omega} \bar{\mathbf{N}}^T \bar{\mathbf{N}} d\Omega + \int_{\Omega} \bar{\mathbf{B}}^T c \bar{\mathbf{B}} d\Omega \right) \bar{\boldsymbol{\varepsilon}}_{eq} = \int_{\Omega} \bar{\mathbf{N}}^T \varepsilon_{eq} d\Omega. \quad (22)$$

The equivalent nodal force vector admissible variation of $\bar{\boldsymbol{\varepsilon}}_{eq}$ can be written as,

$$\mathbf{f}_\varepsilon^{int} = \mathbf{f}_\varepsilon - \left(\int_{\Omega} \bar{\mathbf{N}}^T \bar{\mathbf{N}} d\Omega + \int_{\Omega} \bar{\mathbf{B}}^T c \bar{\mathbf{B}} d\Omega \right) \bar{\boldsymbol{\varepsilon}}_{eq}, \quad \mathbf{f}_\varepsilon = \int_{\Omega} \bar{\mathbf{N}}^T \varepsilon_{eq} d\Omega. \quad (23)$$

The external nodal force vector due to applied boundary conditions is zero ($\mathbf{f}_\varepsilon^{ext} = 0$).

2.3.3. Linearized weak equation

The procedure to linearize field variables and weak equations have been adopted from Geers et al. [\[11\]](#). The linearization of the field variables at iteration i with respect to the previous iteration ($i-1$) at the integration point yields,

$$\tilde{\mathbf{u}}_u^i = \tilde{\mathbf{u}}_u^{i-1} + \delta \tilde{\mathbf{u}}_u^i, \quad \boldsymbol{\varepsilon}^i = \boldsymbol{\varepsilon}^{i-1} + \delta \boldsymbol{\varepsilon}^i, \quad \boldsymbol{\sigma}^i = \boldsymbol{\sigma}^{i-1} + \delta \boldsymbol{\sigma}^i, \quad (24a)$$

$$D^i = D^{i-1} + \delta D^i, \quad \bar{\boldsymbol{\varepsilon}}_{eq}^i = \bar{\boldsymbol{\varepsilon}}_{eq}^{i-1} + \delta \bar{\boldsymbol{\varepsilon}}_{eq}^i, \quad (24b)$$

$$\mathbf{f}_\varepsilon^i = \mathbf{f}_\varepsilon^{i-1} + \delta \mathbf{f}_\varepsilon^i, \quad (24c)$$

with the field variables in the variational form are as follows,

$$\delta \boldsymbol{\varepsilon}^i = \mathbf{B}_u \delta \tilde{\mathbf{u}}_u^i, \quad (25a)$$

$$\delta D^i = \left(\frac{\partial D}{\partial \kappa} \right)^{i-1} \delta \bar{\boldsymbol{\varepsilon}}_{eq}^i = \left(\frac{\partial D}{\partial \kappa} \right)^{i-1} \bar{\mathbf{N}} \delta \bar{\boldsymbol{\varepsilon}}_{eq}^i, \quad (25b)$$

$$\delta \boldsymbol{\sigma}^i = (1 - D^{i-1}) \mathbf{C} \mathbf{B}_u \delta \tilde{\mathbf{u}}_u^i - \mathbf{C} \boldsymbol{\varepsilon}^{i-1} \left(\frac{\partial D}{\partial \kappa} \right)^{i-1} \bar{\mathbf{N}} \delta \bar{\boldsymbol{\varepsilon}}_{eq}^i, \quad (25c)$$

$$\delta \varepsilon_{eq}^i = \left(\frac{\partial \varepsilon_{eq}}{\partial \boldsymbol{\varepsilon}} \right)^{i-1} \delta \boldsymbol{\varepsilon}^i = \left(\frac{\partial \varepsilon_{eq}}{\partial \boldsymbol{\varepsilon}} \right)^{i-1} \mathbf{B}_u \delta \tilde{\mathbf{u}}_u^i, \quad (25d)$$

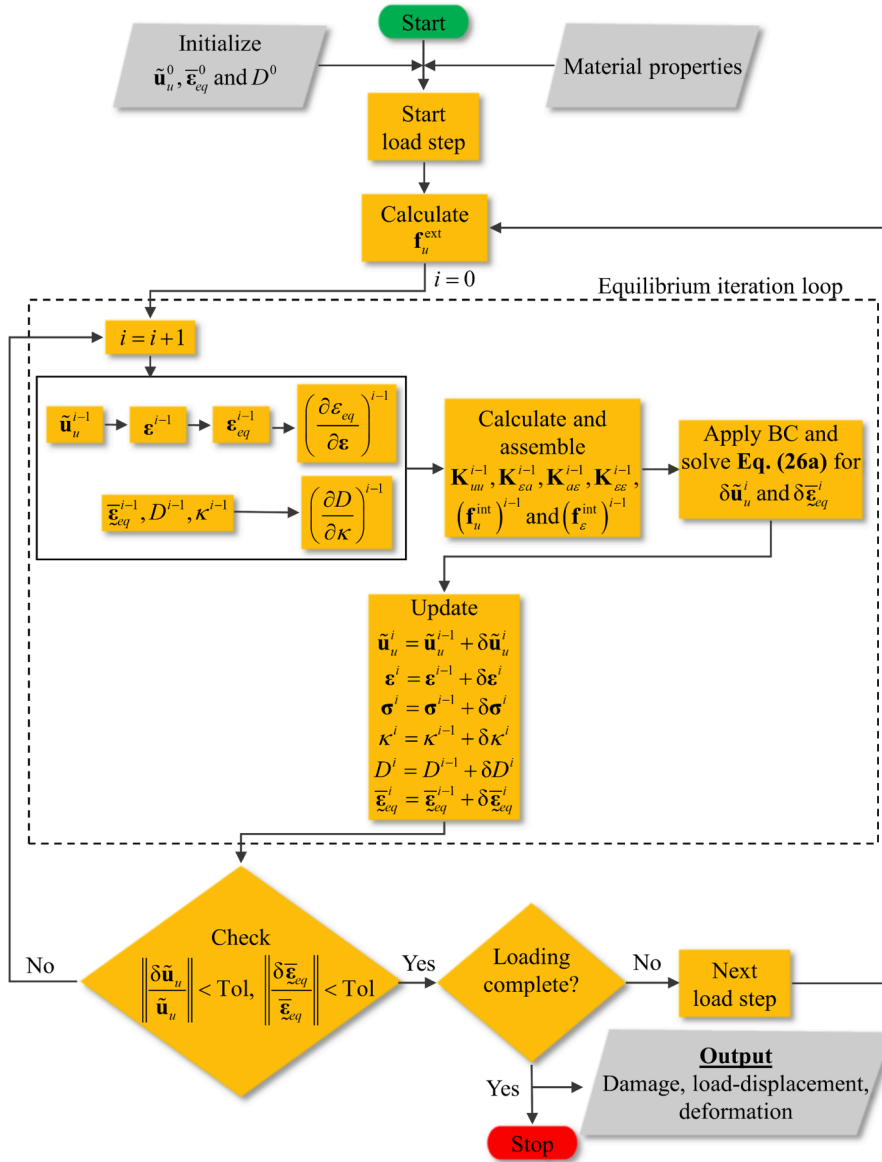


Fig. 2. A flowchart showing the numerical implementation of the proposed XFEM-strain gradient damage model.

$$\delta \mathbf{f}_\varepsilon^i = \left(\frac{\partial \mathbf{f}_\varepsilon}{\partial \varepsilon} \right)^{i-1} \delta \varepsilon^i = \left(\int_{\Omega} \bar{\mathbf{N}}^T \left(\frac{\partial \varepsilon_{eq}}{\partial \varepsilon} \right)^{i-1} \mathbf{B}_u d\Omega \right) \delta \tilde{\mathbf{u}}_u^i, \quad (25e)$$

where details regarding the derivatives of equivalent strain in the matrix, $\frac{\partial \varepsilon_{eq}}{\partial \varepsilon} = \left\{ \frac{\partial \varepsilon_{eq}}{\partial \varepsilon_{xx}} \quad \frac{\partial \varepsilon_{eq}}{\partial \varepsilon_{yy}} \quad \frac{\partial \varepsilon_{eq}}{\partial \gamma_{xy}} \right\}^T$ are mentioned in [Appendix A](#).

In the present study, coupling only between the standard degrees of freedom with local equivalent strain is considered. For a consistent incremental-iterative Newton-Raphson procedure, the discretized weak form of governing equations mentioned in Eq. (20) and Eq. (22) are linearized using Eqs. (24a), (24b), (24c), (25a), (25b), (25c), (25d) and (25e). The global system of equations at iteration i can be written as,

$$\begin{bmatrix} \mathbf{K}_{uu}^{i-1} & \mathbf{K}_{ue}^{i-1} \\ \mathbf{K}_{eu}^{i-1} & \mathbf{K}_{ee}^{i-1} \end{bmatrix} \begin{Bmatrix} \delta \tilde{\mathbf{u}}_u^i \\ \delta \tilde{\varepsilon}_{eq}^i \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}_u^{\text{ext}} \\ 0 \end{Bmatrix} - \begin{Bmatrix} \mathbf{f}_u^{\text{int}} \\ \mathbf{f}_e^{\text{int}} \end{Bmatrix}^{i-1}, \quad (26a)$$

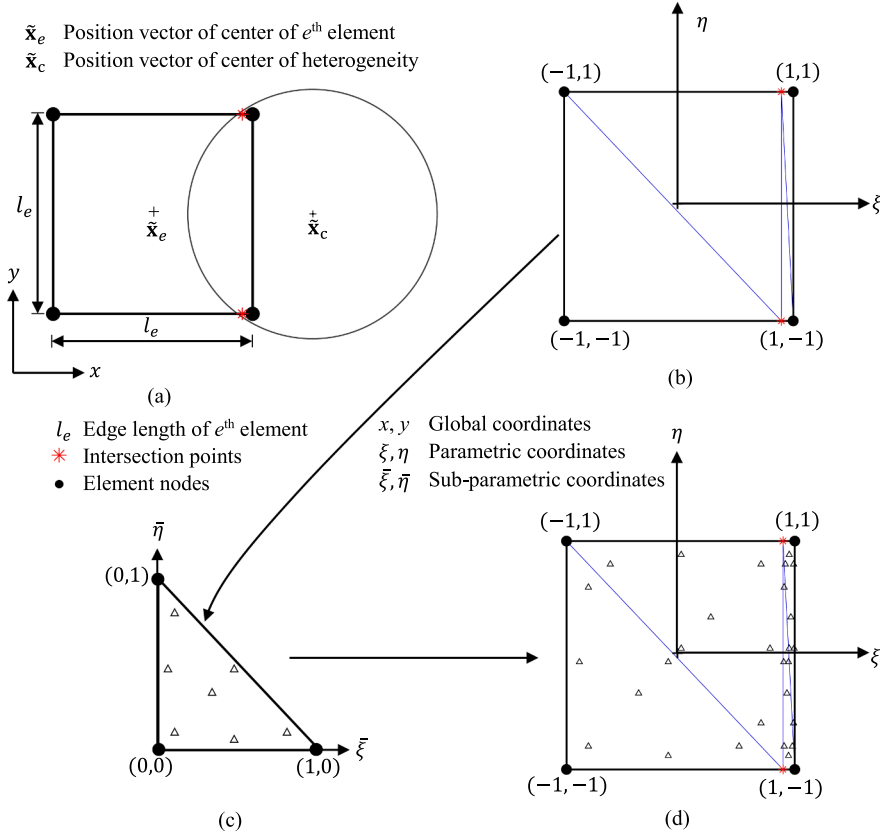


Fig. 3. A schematic of the split element illustrating the numerical integration process along with (a) identification of intersection points, (b) sub-triangulation of element, (c) Gauss points distribution, and (d) mapping of Gauss points.

$$\begin{bmatrix} \mathbf{K}_{aa}^{i-1} & \mathbf{K}_{ab}^{i-1} & \mathbf{K}_{ae}^{i-1} \\ \mathbf{K}_{ab}^{i-1} & \mathbf{K}_{bb}^{i-1} & 0 \\ \mathbf{K}_{ea}^{i-1} & 0 & \mathbf{K}_{ee}^{i-1} \end{bmatrix} \begin{Bmatrix} \delta \tilde{\mathbf{u}}_a^i \\ \delta \tilde{\mathbf{u}}_b^i \\ \delta \tilde{\boldsymbol{\epsilon}}_{eq}^i \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}_a^{\text{ext}} - (\mathbf{f}_a^{\text{int}})^{i-1} \\ \mathbf{f}_b^{\text{ext}} - (\mathbf{f}_b^{\text{int}})^{i-1} \\ \mathbf{f}_e^{i-1} - \mathbf{K}_{ee}^{i-1} \tilde{\boldsymbol{\epsilon}}_{eq}^{i-1} \end{Bmatrix}, \quad (26b)$$

where

$$\mathbf{K}_{aa}^{i-1} = \int_{\Omega} [\mathbf{B}_a]^T (1 - D^{i-1}) \mathbf{C} \mathbf{B}_a d\Omega, \quad (27a)$$

$$\mathbf{K}_{ab}^{i-1} = \int_{\Omega} [\mathbf{B}_a]^T (1 - D^{i-1}) \mathbf{C} \mathbf{B}_b d\Omega, \quad (27b)$$

$$\mathbf{K}_{bb}^{i-1} = \int_{\Omega} [\mathbf{B}_b]^T (1 - D^{i-1}) \mathbf{C} \mathbf{B}_b d\Omega, \quad (27c)$$

$$\mathbf{K}_{ae}^{i-1} = - \int_{\Omega} [\mathbf{B}_a]^T \mathbf{C} \boldsymbol{\epsilon}^{i-1} \left(\frac{\partial D}{\partial \kappa} \right)^{i-1} \bar{\mathbf{N}} d\Omega, \quad (27d)$$

$$\mathbf{K}_{ea}^{i-1} = \int_{\Omega} \bar{\mathbf{N}}^T \left(\frac{\partial \boldsymbol{\epsilon}_{eq}}{\partial \boldsymbol{\epsilon}} \right)^{i-1} \mathbf{B}_a d\Omega, \quad (27c)$$

$$\mathbf{K}_{ee}^{i-1} = \left(\int_{\Omega} \bar{\mathbf{N}}^T \bar{\mathbf{N}} d\Omega + \int_{\Omega} \bar{\mathbf{B}}^T \mathbf{C} \bar{\mathbf{B}} d\Omega \right), \quad (27d)$$

$$(\mathbf{f}_a^{\text{int}})^{i-1} = \int_{\Omega} \mathbf{B}_a^T \boldsymbol{\sigma}^{i-1} d\Omega, \quad (27e)$$

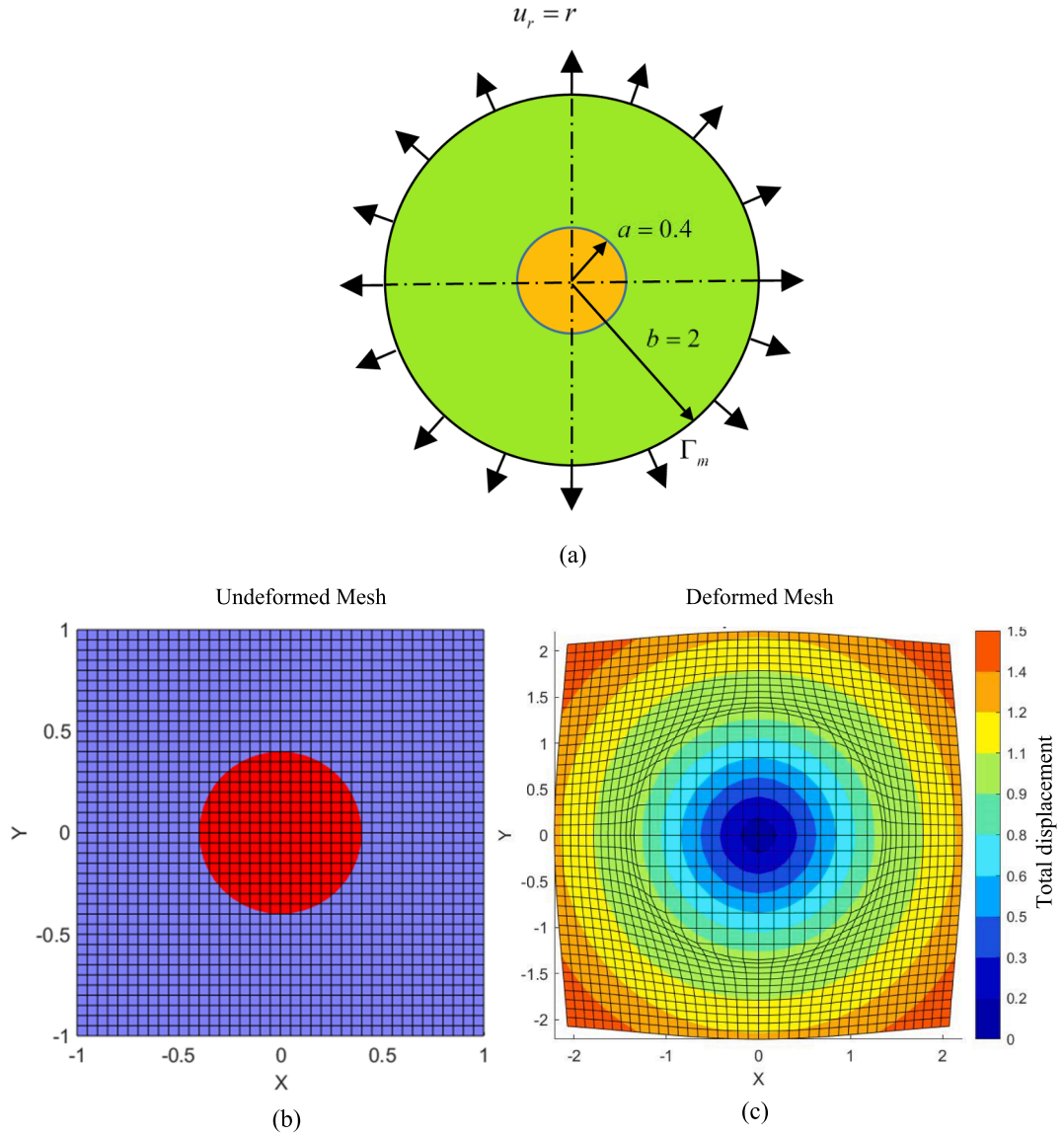


Fig. 4. A schematic of (a) bi-material boundary value problem consisting of the soft reinforced particle with radius 0.4 unit and radial displacement as a boundary condition (b) undeformed and (c) deformed square domain meshed with quadrilateral elements.

$$(\mathbf{f}_b^{int})^{i-1} = \int_{\Omega} \mathbf{B}_b^T \boldsymbol{\sigma}^{i-1} d\Omega, \quad (27f)$$

$$\mathbf{f}_a^{ext} = \int_{\Gamma} \mathbf{N}_a^T \mathbf{t} d\Gamma, \quad (27g)$$

$$\mathbf{f}_b^{ext} = \int_{\Gamma} \mathbf{N}_b^T \mathbf{t} d\Gamma, \quad (27h)$$

$$\mathbf{f}_\varepsilon^{i-1} = \int_{\Omega} \tilde{\mathbf{N}}^T \varepsilon_{eq}^{i-1} d\Omega. \quad (27i)$$

The numerical implementation of the proposed XFEM-strain gradient damage model is shown using flowchart in Fig. 2.

2.4. Numerical integration

In this section, the numerical integration scheme used in the proposed XFEM-strain gradient damage model has been presented. The discretized domain contains non-split elements (standard elements) and split elements (enriched elements). A quadratic eight-node quadrilateral element (Q8) is used to approximate displacement and a linear four-node quadrilateral element (Q4) is used to approximate local equivalent strain. For standard, a three-point Gauss quadrature rule (total nine Gauss points in an elemental domain) is used. The implementation of standard Gauss quadrature over split element (element intersected by heterogeneity) results in an inaccurate integration of enrichment functions [35].

For such a split-element, the element partitioning scheme is used. The split element is partitioned into sub-elements (sub-triangles) depending upon the number and location of the intersection points. The procedure for implementation of the numerical integration scheme based on sub-triangulation is as follows,

- (a) Determine the location of intersection points between the element boundary and the heterogeneity boundary as shown in Fig. 3(a).
- (b) The element is partitioned into sub-triangles as shown in Fig. 3(b).
- (c) Seven Gauss points are distributed in each triangle shape as shown in Fig. 3(c).
- (d) Gauss points with their respective weights from all the sub-triangles are mapped back to the element domain as shown in Fig. 3(d).

2.4.1. Compatibility analysis

The purpose of the analysis is to check the compatibility of ridge enrichment function and numerical integration scheme with quadratic element. The bi-material boundary value problem is solved using XFEM at different mesh densities. The numerically obtained displacement field is compared with analytical solution. The schematic of bi-material boundary value problem is shown in Fig. 4(a).

By imposing linear displacement field $u_r = r$ and $u_\theta = 0$ on the matrix boundary Γ , the analytical displacement solution of the bi-material problem is given by,

$$u_r^A = \begin{cases} \left[\left(1 - \frac{b^2}{a^2} \right) \alpha + \frac{b^2}{a^2} \right] r & 0 \leq r \leq a, \\ \left(r - \frac{b^2}{r} \right) \alpha + \frac{b^2}{r} & a \leq r \leq b, \end{cases} \quad (28)$$

$$u_\theta^A = 0, \quad (29)$$

$$\alpha = \frac{(\lambda_1 + \mu_1 + \mu_2)b^2}{(\lambda_2 + \mu_2)a^2 + (\lambda_1 + \mu_1)(b^2 - a^2) + \mu_2b^2}, \quad (30)$$

where $\lambda_1 = \frac{E_1\nu_1}{(1+\nu_1)(1-2\nu_1)}$, $\lambda_2 = \frac{E_2\nu_2}{(1+\nu_2)(1-2\nu_2)}$ are Lamé constants and $\mu_1 = \frac{E_1}{2(1+\nu_1)}$, $\mu_2 = \frac{E_2}{2(1+\nu_2)}$ are shear modulus of soft inclusion and matrix respectively. The material constants E_1 , E_2 , ν_1 and ν_2 are assumed to be 1 unit, 10 unit, 0.25 and 0.3 respectively. The geometric constants b and a are assumed to be 2 and 0.4 respectively. Fig. 4(b) shows the square domain of length equal to 2 unit meshed with quadrilateral elements for the numerical analysis. Fig. 4(c) shows the deformed mesh after the displacement conditions $u_r = r$ and $u_\theta = 0$ are imposed over the boundary of a square domain. The numerical analysis is performed for two different XFEM cases as follows,

- (a) XFEM with 4-noded quadrilateral element
- (b) XFEM with 8-noded quadrilateral element

For Case(a), the XFEM split element consists of 4-standard and 4-enriched degrees of freedom. For Case(b), the XFEM split element consists of 8-standard and 8-enriched degrees of freedom. Analysis was performed for four different mesh densities i.e., 21×21 , 31×31 , 41×41 , and 61×61 elements. The L2 norm is used as an error estimator for the numerical displacement field. The expression of the L2 norm is as follows,

$$L^2(u) = \sqrt{\frac{\sum_{j \in I} \|\tilde{u}_j - u_j^A\|^2}{N}}, \quad (31)$$

where $\tilde{u}_j$ is the displacement vector at j^{th} node obtained using XFEM. u_j^A is the analytical displacement vector at j^{th} node obtained from the analytical displacement solution in Eq. (28) and Eq. (29). Although convergence order of the quadratic element is less than linear element for XFEM but the value of L2 norm for both linear and quadratic element is same for dense mesh as shown in Fig. 5.

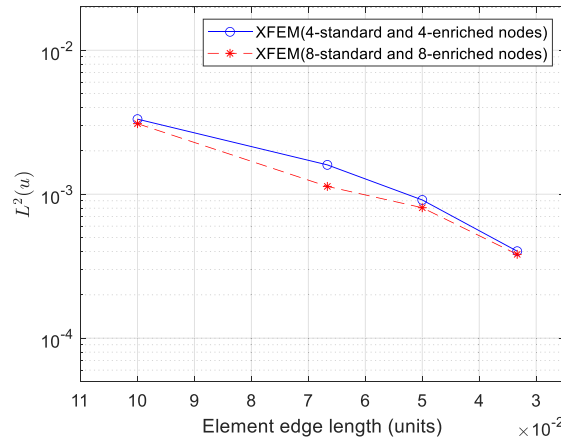


Fig. 5. A variation of L2 norm with decrease in element edge length (increase in mesh density) for a square domain solved using linear (4-standard and 4-enriched nodes) and quadratic (8-standard and 8-enriched nodes) elements.

3. Numerical results and discussions

In this section, capability of XFEM-strain gradient model has been demonstrated by simulating mode-I, mode-II, and mixed mode failure mechanism using SENB, SEN and L-shaped specimen with reinforcement (hard and soft) particles respectively. Mesh-sensitivity and particle-sensitivity analysis were also performed to quantify effect of various parameters like mesh size, particle size and volume fraction on behavior of SENB specimen.

The numerical simulations were performed based on following assumptions and simplifications,

1. Damage properties of the matrix were calibrated using SENB specimen to match the experiments.
2. A perfect bond between the matrix and reinforcement particles was assumed with no interphase zone.
3. The elastic and damage properties of the hard and soft reinforcement were assumed by considering the properties of the matrix as baseline.
4. Displacement control simulations were performed under plane strain condition.
5. Mechanical properties of the matrix and reinforcement particles were kept the same for all the problems.
6. The dimensions of the single edge notched specimen and L-shape specimen were calibrated to suite the gradient parameter (c) and damage evolution parameter (β).

3.1. Mode-I Problem: Single edge notch beam (SENB) specimen

In this section, the mode-1 failure mechanism in SENB specimen with reinforcement particles has been demonstrated using proposed model (XFEM-strain gradient damage model). The simulation of SENB with reinforcement particles requires mechanical properties (elastic constants and damage parameters) of both matrix and reinforcement particle as input to the proposed model. Therefore, first the mechanical properties of matrix were obtained by verifying with the experimental data available in the literature using the standard model (FEM-strain gradient damage model). Then, the mechanical properties of the soft and hard reinforcement particle were assumed by considering the properties of matrix as baseline. After identifying the mechanical properties of both matrix and reinforcement particles, the following analysis were performed with specific objectives as listed below,

Mesh-sensitivity analysis of SENB specimen with soft and hard reinforcement particles,

- To verify solution obtained from the proposed model with the standard model.
- To identify computational performance of the proposed model.

Particle sensitivity analysis of SENB specimen with soft and hard reinforcements particles were performed,

- To quantify effect of particle size and volume fraction on the peak load in response curve.

3.1.1. Matrix property optimization

The mechanical properties of matrix were optimized using SENB specimen without reinforcement particles. The geometric details of SENB specimen along with boundary condition is shown in Fig. 6(a). Domain of SENB specimen was discretized with eight noded quadratic quadrilateral elements (Q8) as shown in Fig. 6(b). The thickness of SENB specimen is 13 mm. SENB specimen was simulated under plane strain condition using FEM-strain gradient damage model. The elastic properties of matrix were taken from the literature [2]. The gradient parameter (c) and the damage evolution parameter (β) greatly influences load–displacement curve. Hence, all the

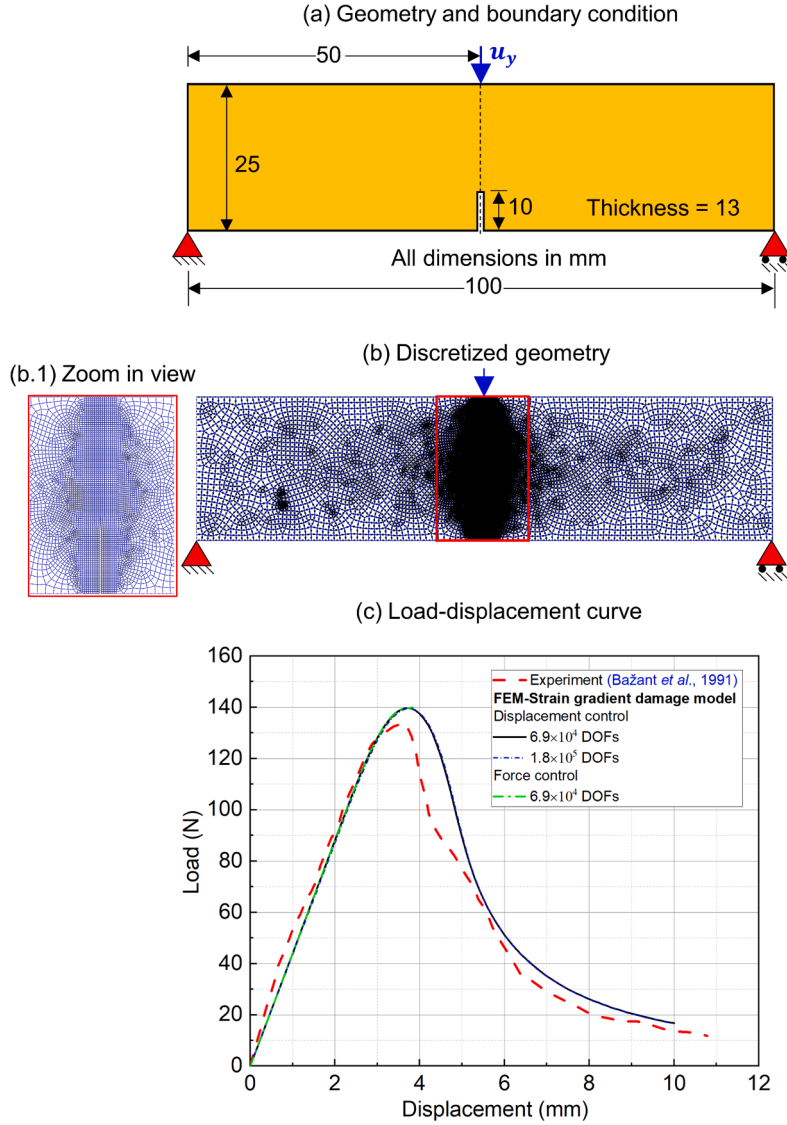
Matrix Property Optimization: SENB specimen without reinforcement particles

Fig. 6. (a) A schematic representation of SENB specimen with geometric dimensions and boundary conditions used for optimizing damage parameters of the matrix. (b) Discretized SENB specimen geometry with Q8 elements. (c) Comparison of simulated response curves with the experimental response curve.

Table 1

Mechanical properties of matrix.

Elastic properties [2]	Damage properties (Optimized)
$E = 15.3 \text{ GPa}$, $\nu = 0.15$	$k_r = 8$, $\alpha = 0.99$, $\beta = 600$, $c = 1$, $\kappa_0 = 4.3 \times 10^{-4}$

damage parameters for the matrix were calibrated to match the experimental load–displacement curve [2]. The optimized values of matrix damage parameters along with elastic constants are tabulated in Table 1. The SENB response curve was obtained by estimating reaction load after gradually applying the transverse displacement at center point of the beam. The SENB specimen is solved for two different mesh densities (with 9785 and 25,579 Q8 elements) using displacement control scheme and one mesh density (with 9785 Q8 elements) using force control scheme. The converged solution was obtained till the peak load using force control scheme. The damage parameters were tuned to match the numerical response curve with the experiments using displacement control scheme as shown in Fig. 6(c). The numerically obtained response curves using displacement control scheme are insensitive to the mesh size.

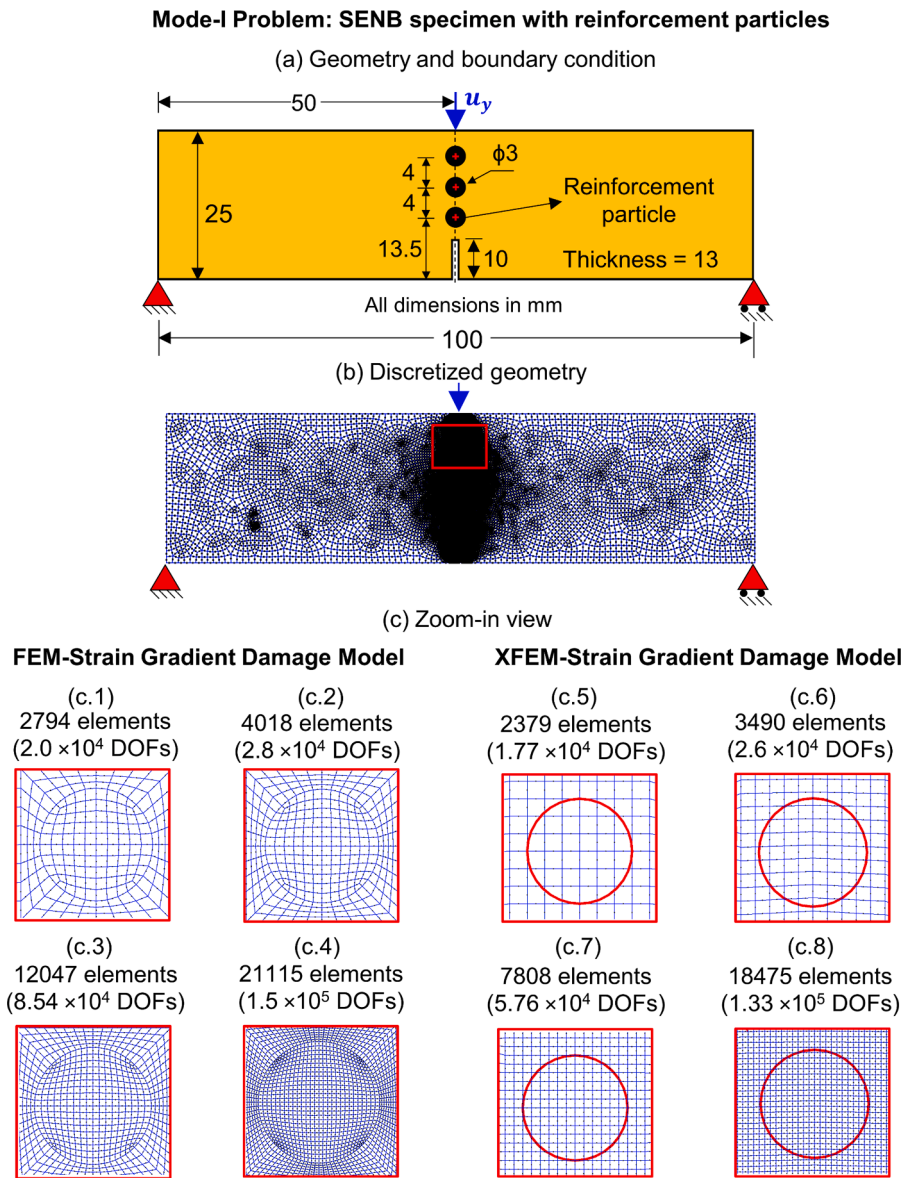


Fig. 7. (a) A schematic representation of SENB specimen with geometric dimensions, spatial distribution of reinforcement particles and boundary conditions used for mesh sensitivity analysis. (b) Discretized SENB specimen geometry using Q8 elements with (c) zoom-in view of four conformal mesh densities for FEM-strain gradient damage model and four non-conformal mesh densities for XFEM-strain gradient damage model.

Table 2
Assumed material properties soft and hard reinforcement particles.

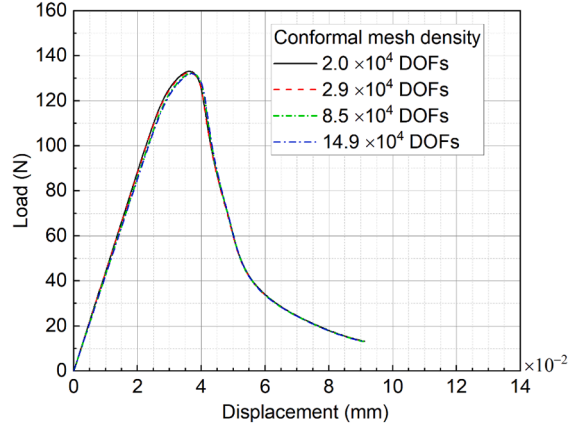
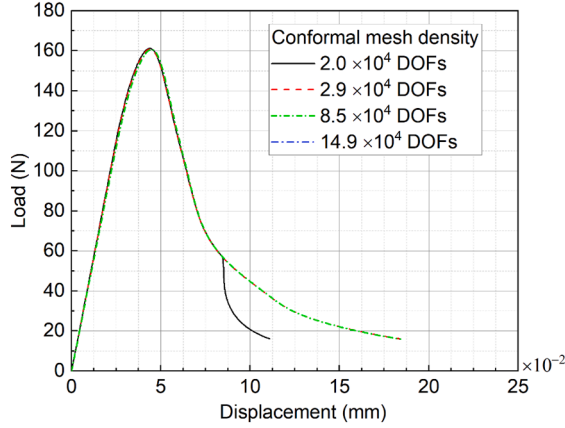
Material parameters	Reinforcement Particles	
	Hard	Soft
E	30 GPa	5 GPa
ν	0.15	0.15
k_r	10	10
α	0.99	0.99
β	800	800
c	2	2
κ_0	8×10^{-4}	8×10^{-4}

Mesh Sensitivity analysis : Variation of load-displacement curve

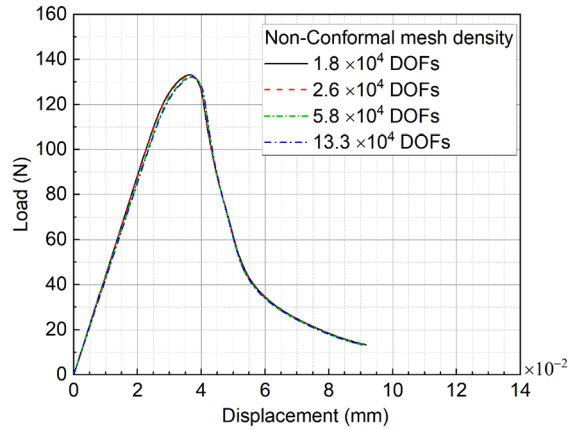
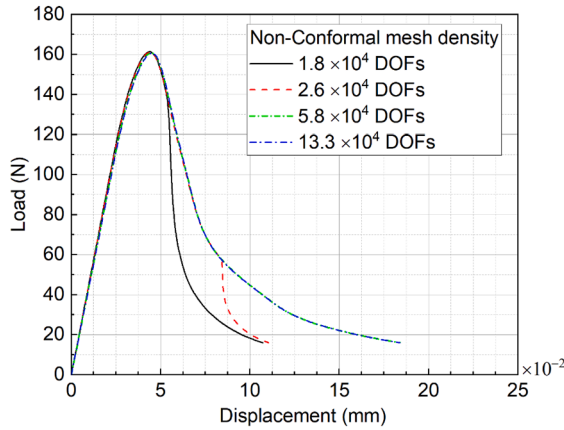
Hard reinforcement particles

Soft reinforcement particles

(a) FEM-Strain Gradient Damage Model



(b) XFEM-Strain Gradient Damage Model



(c) Variation of peak load and area under the response curve

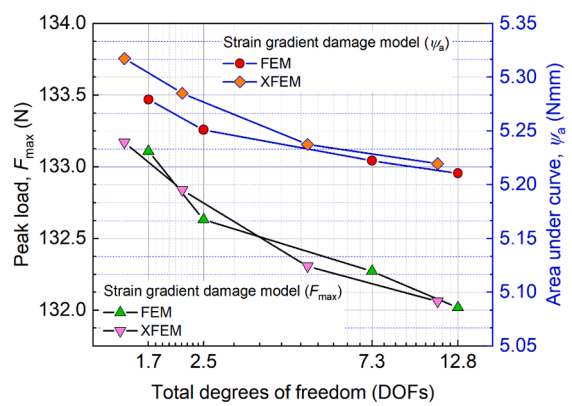
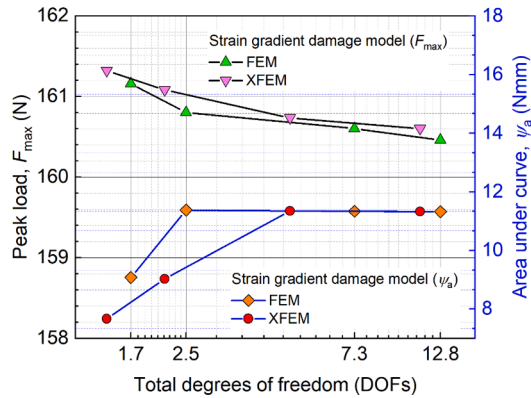


Fig. 8. (a) A load–displacement response curve for SENB specimen with hard (left column) and soft (right column) reinforcement particles obtained using standard FEM model at different mesh densities. (b) A load–displacement response curve for SENB specimen with hard (left column) and soft (right column) reinforcement particles obtained using proposed XFEM model at different mesh densities. (c) Variation of peak load and area under curve for SENB specimen with hard reinforcement particles (left) and soft reinforcement (right) for both standard FEM and proposed XFEM model at different mesh densities.

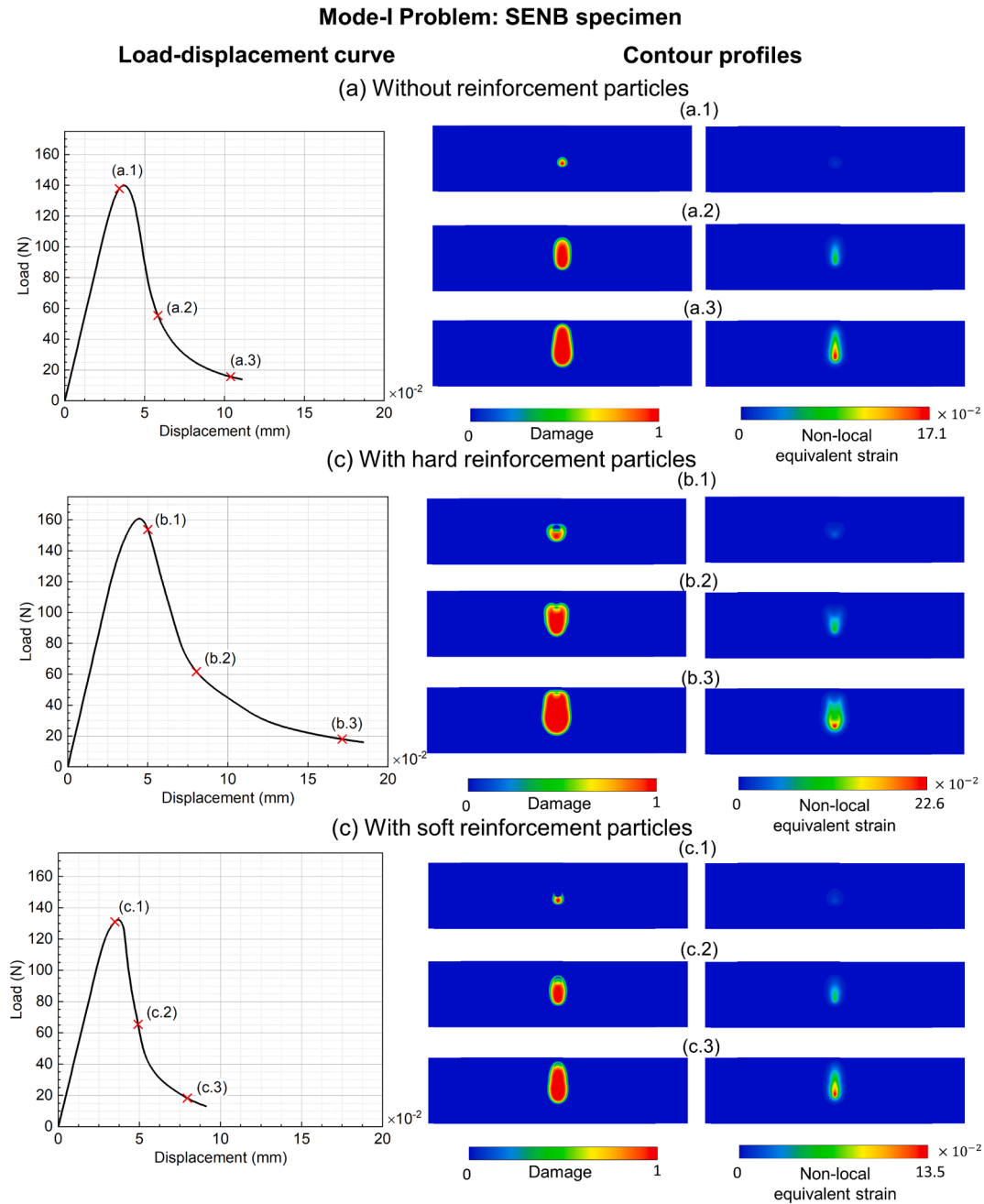


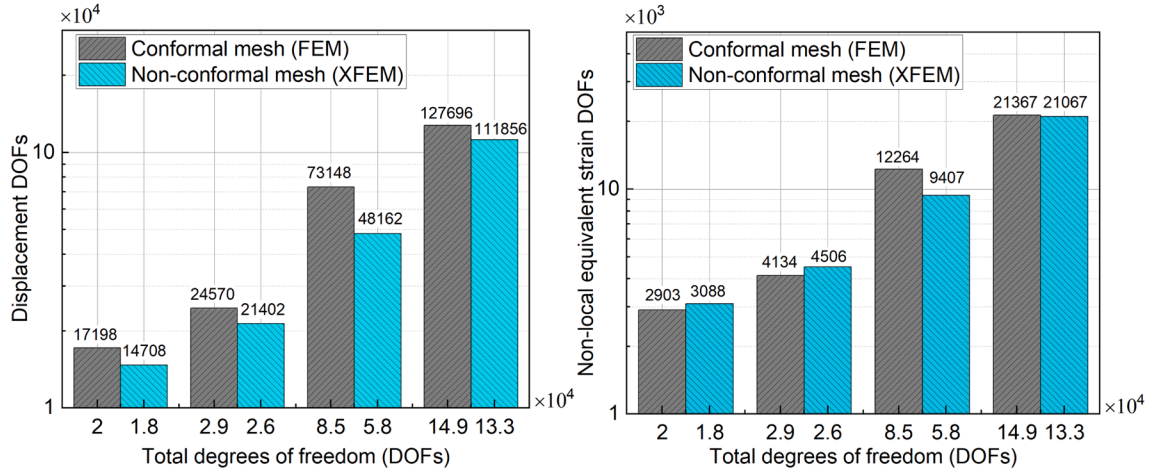
Fig. 9. The load–displacement curve along with contour plots (damage and non-local equivalent strain evolution) for SENB specimen for three different cases (a) without reinforcement particle (b) with hard reinforcement particle and (c) with soft reinforcement at three loading stages.

3.1.2. Mesh-Sensitivity analysis

The mesh sensitivity analysis of SENB specimen with reinforcement particles (soft and hard) was performed using proposed and standard model. The details of SENB geometry, spatial distribution of reinforcement particles and boundary conditions are shown in Fig. 7(a). The location of reinforcement particles was kept ahead of crack tip to impose maximum effect on load–displacement curve during simulations. The thickness of SENB specimen is 13 mm. The SENB specimen domain was discretized with Q8 elements. The displacement control solution was implemented under plane strain condition. Fig. 7(b) shows four different conformal and non-conformal mesh densities generated for implementing FEM-strain gradient damage model (standard) and XFEM-strain gradient damage model (proposed) respectively. The mechanical properties of matrix and reinforcement particles used during the analysis are tabulated in Tables 1 and 2 respectively.

Computational Performance: SENB specimen with reinforcement particles

(a) Composition of the degree of freedoms



(b) Total computation time

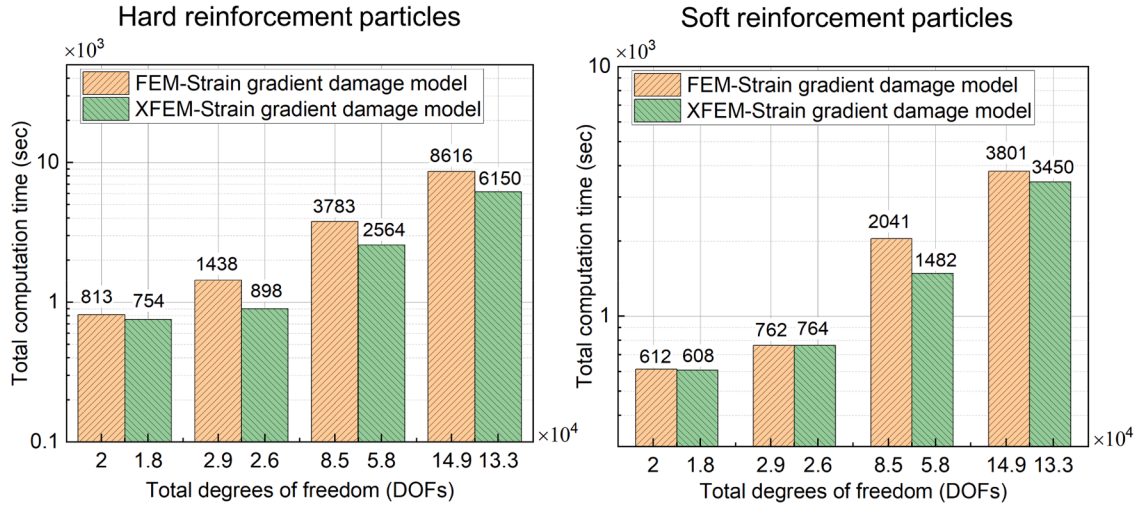


Fig. 10. (a) A comparison between number of displacement (left) and non-local equivalent strain DOFs (right) at four conformal (FEM-strain gradient damage model) and non-conformal mesh densities (XFEM-strain gradient damage model). (b) A histogram for comparison between the computational time required by standard FEM and proposed XFEM model for solving SENB specimen with hard (left) and soft (right) reinforcement particles at different mesh densities.

3.1.2.1. Variation in load–displacement curve. For SENB specimen with hard reinforcement particles, the load–displacement plots obtained using the standard and proposed model are shown in the left column of Fig. 8(a) and (b) respectively. For the first two coarse mesh (Fig. 7(b.5) and (b.6)) of proposed model and the first coarse mesh (Fig. 7(b.1)) of standard model, the SENB specimen fails at approximate 39% lower displacement than all other mesh densities. For effective quantification of differences among the load–displacement curves, the variation of two parameters i.e., maximum load-carrying capacity (F_{max}) and area under the curve (ψ_a) were shown in left column of Fig. 8(c). The overall variation of peak load is less 2% for both standard and proposed model. The area under the curve for SENB specimen with coarse mesh density is 20% lower compared to SENB specimen with high mesh density.

For SENB specimen with soft reinforcement particles, the load–displacement plots obtained using the standard and proposed model are shown in the right column of Fig. 8(a) and (b) respectively. The variation among load–displacement plots is minimal. Fig. 8(c) on right column shows the overall variation of maximum load-carrying capacity and area under the curve is less than 3% for both standard and proposed model.

The effect of reinforcement particles on load–displacement curve for SENB specimen was demonstrated using three different cases i. e., without reinforcement, with hard reinforcement and with soft reinforcement as shown in the Fig. 9. The results were obtained for SENB specimen with 7908 elements (5.76×10^4 DOFs) using XFEM-strain gradient damage model. For SENB specimen without

Particle Sensitivity Analysis: Realizations in SENB Specimen

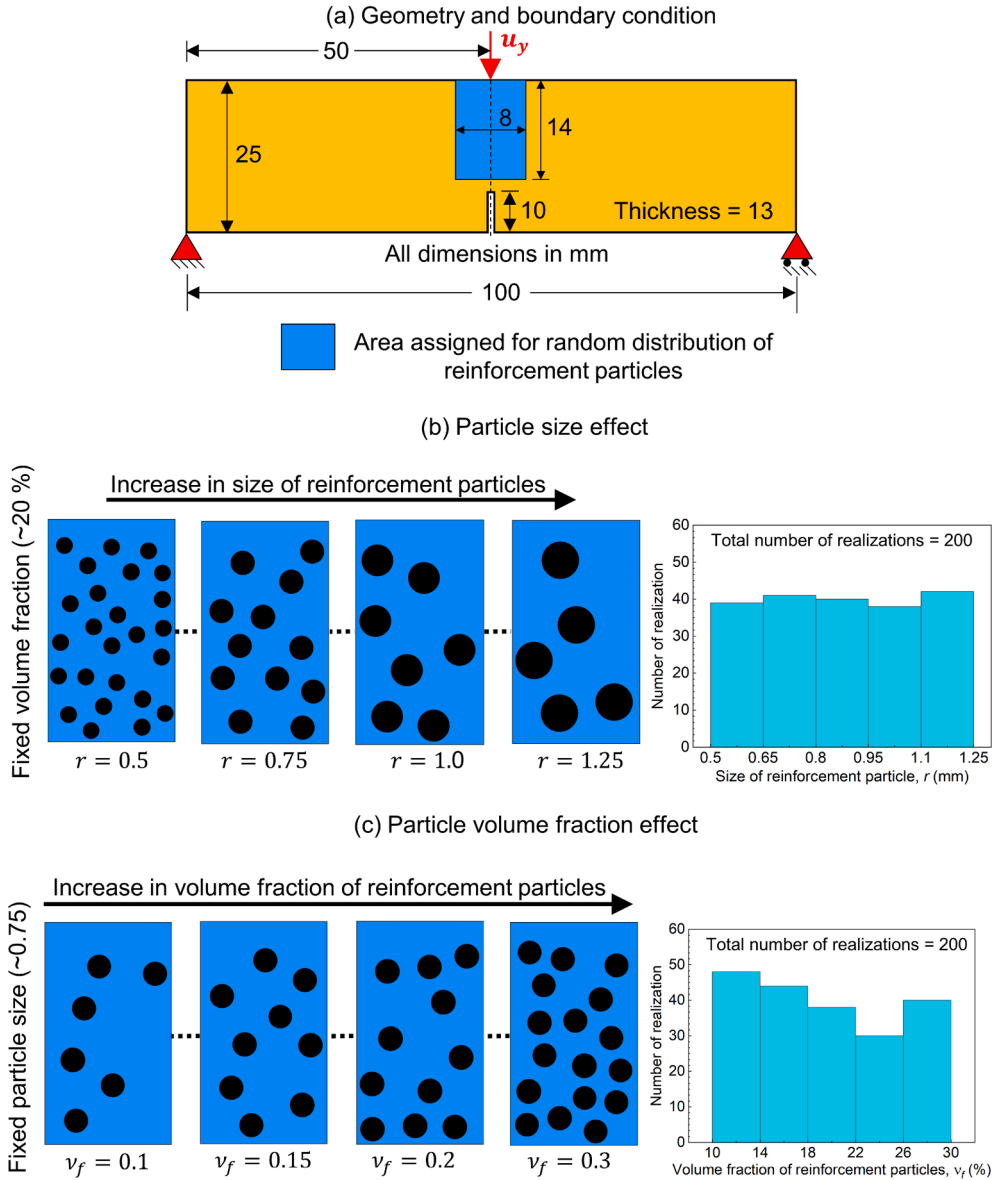


Fig. 11. (a) A schematic representation of SENB specimen with geometric dimensions and boundary conditions used for mesh sensitivity analysis. (b) 200 randomly generated realizations for effect of particle size with fixed volume fraction of 20%. (c) 200 randomly generated realizations for effect of particle volume fraction with fixed particle size of 0.75 mm.

reinforcement particles, the variable in the code that define reinforcement particles was kept empty. The results shows that the elastic modulus of reinforcement particles greatly influences peak load of SENB specimen. The peak load of SENB specimen with hard reinforcement particles is 15% and 18% higher than the load carrying capacity of SENB specimen without reinforcement particles and with soft reinforcement particles respectively. Moreover, the SENB specimen with soft reinforcement particles fails at 10% and 20% lower displacement value than that of SENB specimen without and with hard reinforcement particles respectively. For each case, the damage evolution and non-local equivalent strain contour plot at three different loading stages is shown in Fig. 9.

3.1.2.2. Computational performance. In this section, computational performance of XFEM and FEM strain gradient damage model is compared based on the analysis performed during mesh sensitivity analysis. The computer system used for the present study was equipped with AMD Ryzen Threadripper 3975WX@3.5 GHz (32-cores/64-threads) processor with 128 GB of RAM. The code has been written and simulated in parallel on a MATLAB platform.

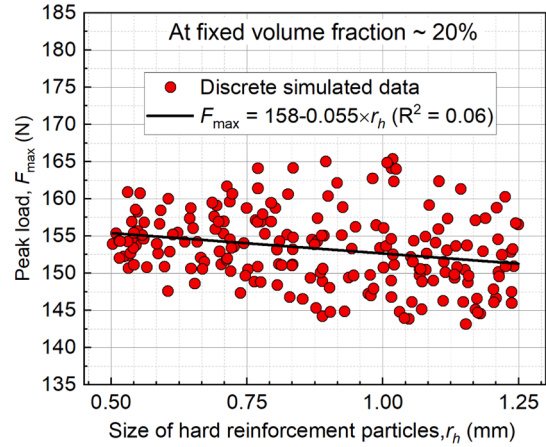
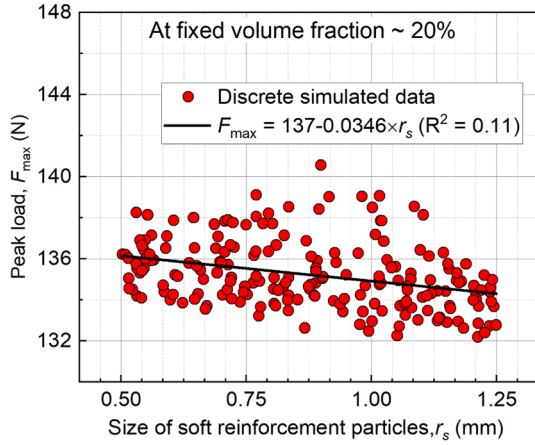
Both FEM and XFEM-strain gradient damage model was solved for displacement and non-local equivalent strain degrees of freedom

Particle Sensitivity Analysis: Effect of Particle Size and Volume Fraction

Hard reinforcement particles

Soft reinforcement particles

(a) Reinforcement particle size



(b) Reinforcement particle volume fraction

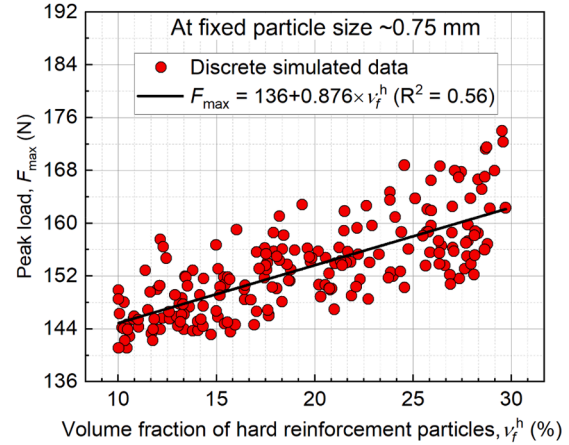
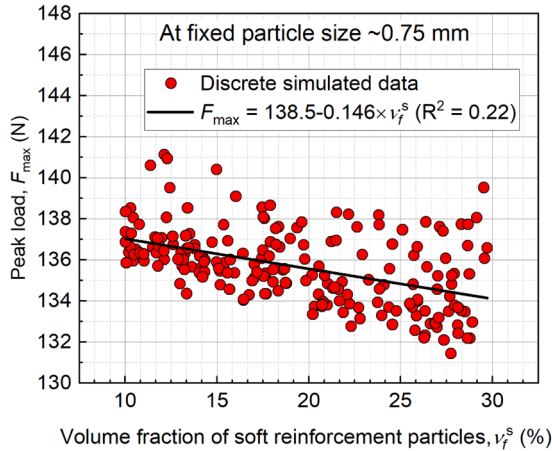


Fig. 12. (a) A correlation plot between size of reinforcement particles (hard particles in left column and soft particles in right column) and peak load of SENB specimen. Each data point in the plot represents a peak load (scaled on Y-axis) obtained after simulating SENB specimen with realization generated for a particular size of reinforcement particles (scaled on X-axis). (b) A correlation plot between volume fraction of reinforcement particles and peak load of SENB specimen.

(DOFs). Total number of displacements DOFs in XFEM is the sum of standard and enriched DOFs. For FEM, number of displacements DOFs is equal to standard DOFs i.e., twice the number of nodes in the discretized geometry. Fig. 10(a) shows the distribution of displacement and non-local equivalent DOFs for XFEM and FEM-strain gradient damage model. The overall difference in total degrees of freedom between conformal and non-conformal mesh is at least 10%.

Total 8 simulations (4 conformal FEM mesh densities + 4 non-conformal XFEM mesh densities) were performed for SENB specimen with hard reinforcement particles. Similarly, total 8 simulations were performed for SENB specimen with soft reinforcement particles. Factors influencing computation time such as number of parallel processors, parallel loops, incremental load steps and equilibrium tolerance limit were kept same for all the simulation. Fig. 10(b) shows a histogram plot with computation time required by FEM and XFEM-strain gradient damage model for solving SENB specimen with hard (left) and soft (right) reinforcement particles at four different mesh densities. For hard reinforcement, XFEM strain gradient damage model is approximately 28% efficient than FEM-strain gradient damage model for SENB specimen with high mesh density. For soft reinforcement, XFEM-strain gradient damage model is approximately 9% efficient for highest mesh density. The computational performance of both models is almost identical for SENB specimen with coarse mesh densities.

XFEM-Strain Gradient Damage Model

Mode-II Problem: Single edge notched specimen

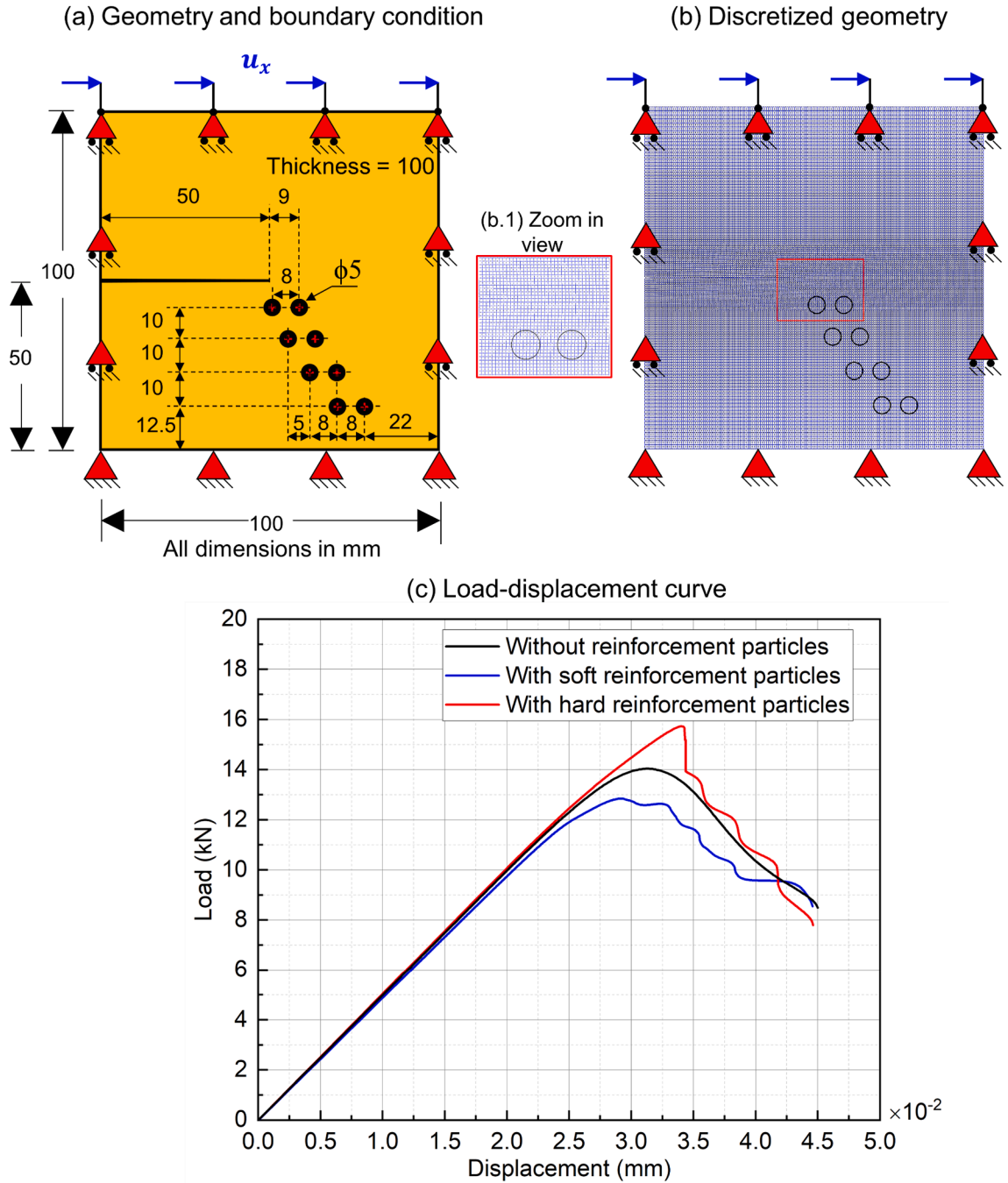


Fig. 13. (a) A schematic representation of single edge notched (SEN) specimen with geometric dimensions, spatial distribution of reinforcement particles and boundary conditions used for simulating mode-II failure mechanism. (b) Discretized geometry of SEN specimen with Q8 elements. (c) Load-displacement curve for SEN specimen with three different cases i.e., without reinforcement particles, with hard reinforcement particles and with soft reinforcement particles.

Mode-II Problem: Single edge notched specimen

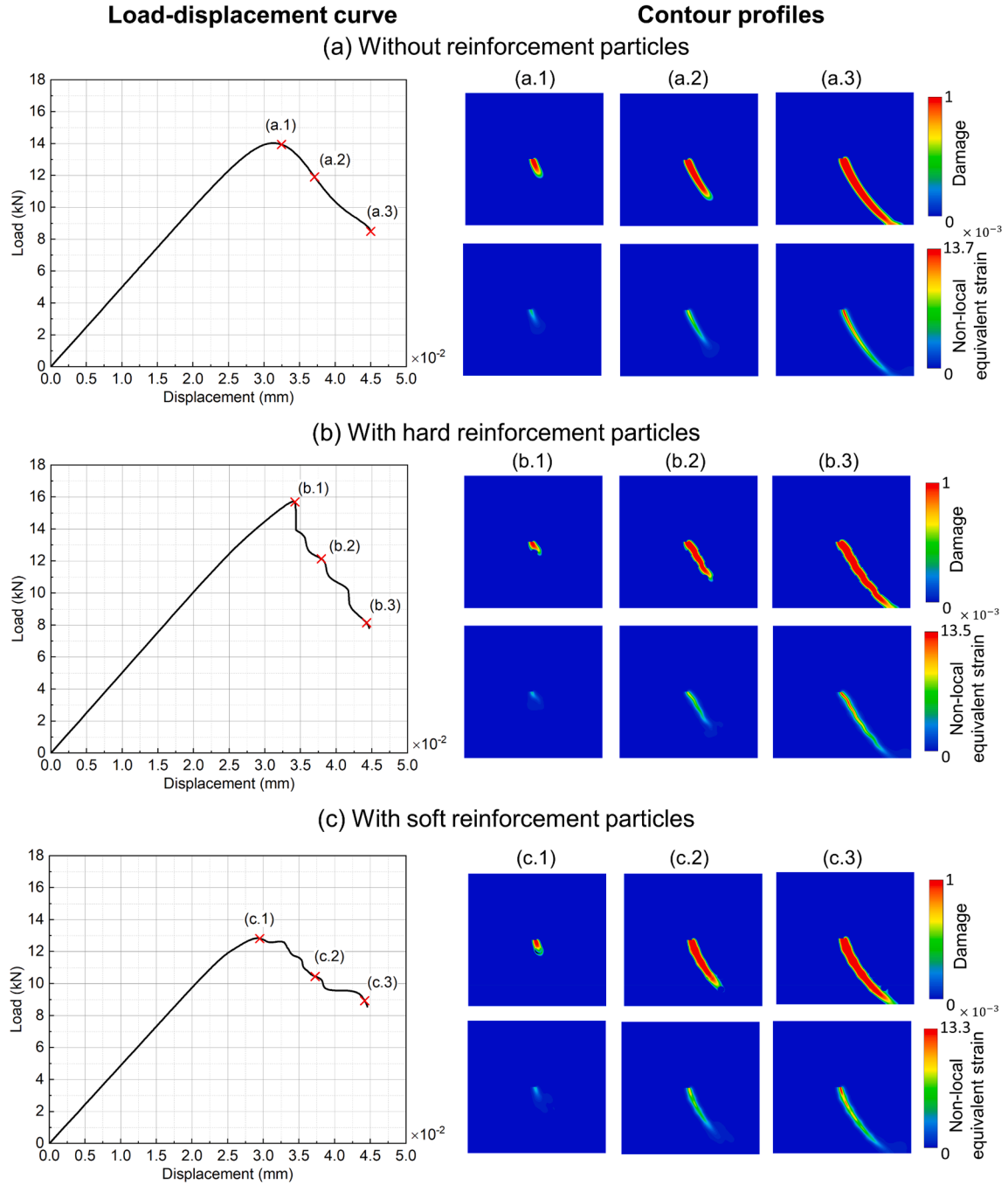


Fig. 14. The load–displacement curve along with contour plots (damage and non-local equivalent strain evolution) for SEN specimen with three different cases (a) without reinforcement particle (b) with hard reinforcement particle and (c) with soft reinforcement at three different loading stages.

XFEM-Strain Gradient Damage Model

Mix Mode Problem: L-shaped specimen

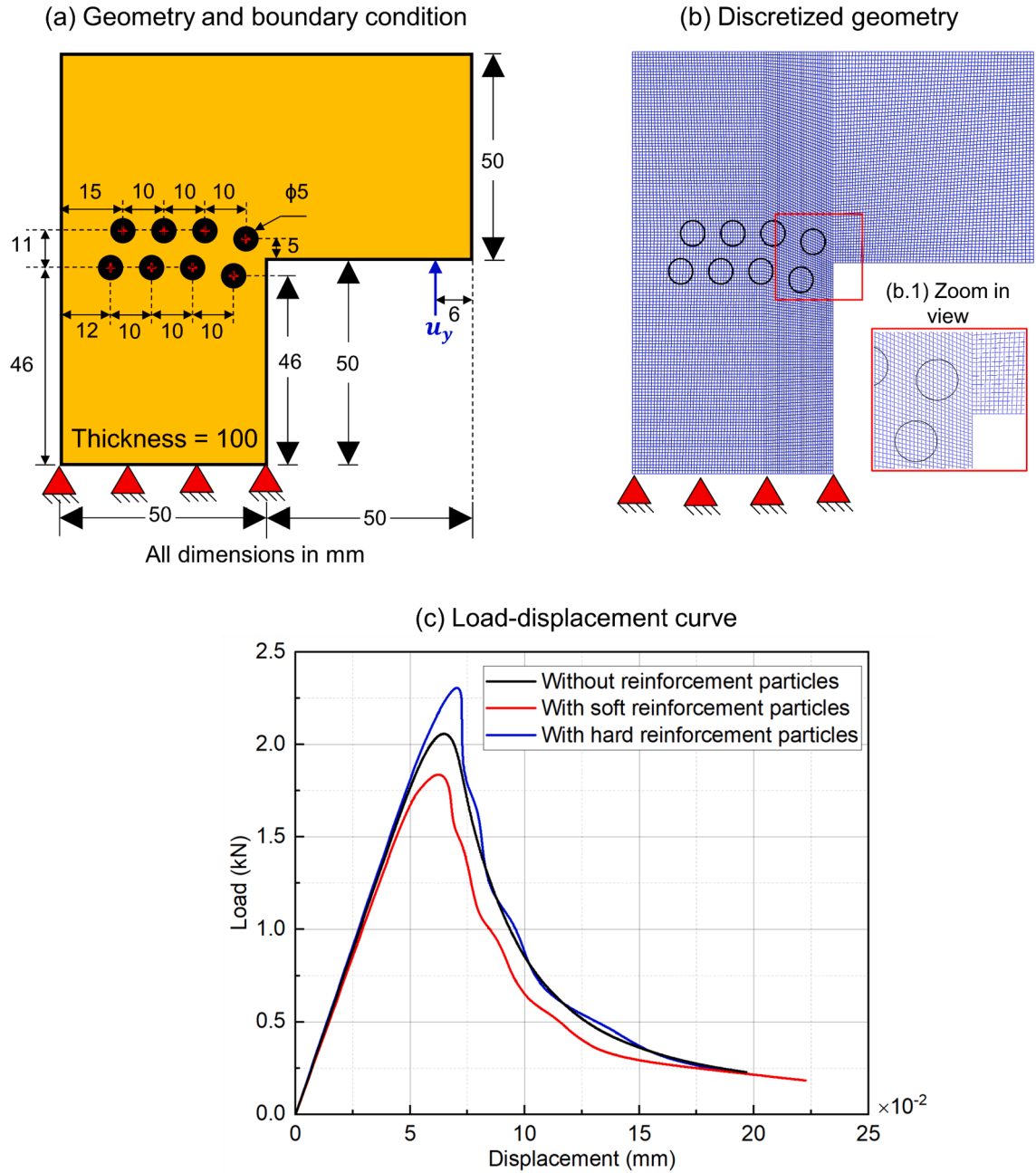


Fig. 15. (a) A schematic representation of L-shaped specimen with geometric dimensions, spatial distribution of reinforcement particles and boundary conditions used for simulating mix-mode failure mechanism. (b) Discretized geometry of L-shaped specimen with Q8 elements. (c) Load-displacement curve for L-shaped specimen with three different cases i.e., without reinforcement particles, with hard reinforcement particles and with soft reinforcement particles.

3.1.3. Particle-sensitivity analysis

In this section, results obtained from particle-sensitivity analysis are presented and discussed. The analysis was performed using XFEM-strain gradient damage model under plane strain condition. The main objective of particle-sensitivity analysis was to quantify,

- Effect of reinforcement particle size on peak load of SENB specimen.
- Effect of reinforcement particle volume fraction on peak load of SENB specimen.

A total of 400 realizations (200 for particle size + 200 for particle volume fraction) with random spatial distribution of reinforcement particle were generated for the analysis.

3.1.3.1. Effect of reinforcement particle size on peak load of SENB specimen. 200 realizations with fixed volume fraction of 20% were generated by randomly varying size (radius) of reinforcement particles from 0.5 to 1.25 mm in the region ahead of crack tip as shown in Fig. 11(a). The realizations were generated in two steps as follows,

- (a) A set of 200 random numbers between 0.5 and 1.25 was created using “rand” inbuilt function in MATLAB. Each random number in the set represents the radius of reinforcement particles. The histogram plot of set of the random numbers is shown in the right of Fig. 11(b).
- (b) Each random number in the set represent radius of reinforcement particles. For each random number, a realization was generated using a size and spatial distribution algorithm mentioned in Bansal et al. [4]. The realizations with different size of reinforcement particles are shown in Fig. 11(b).

Total 400 simulations (200 with hard reinforcement + 200 with soft reinforcement) were performed for SENB specimen with a set of 200 realizations. SENB specimen geometry with 3490 Q8 elements (2.6×10^4 DOFs). The material properties of matrix and reinforcement particles used for the analysis are tabulated in Tables 1 and 2 respectively. To increase computational efficiency, the simulations were terminated after the detection of peak load in SENB specimen. For both hard and soft reinforcement particles, the correlation between size and peak load is negative as shown in Fig. 12(a). For SENB specimen with hard reinforcement particles, the linear fit predicts that the peak load is reduced by 2.7% with an increase in size of hard reinforcement particles from 0.5 mm to 1.25 mm. The peak load of SENB specimen is dropped by 2% with an increase in size of soft reinforcement particles from 0.5 mm to 1.25 mm.

3.1.3.2. Effect of reinforcement particle volume fraction on peak load of SENB specimen. With the fixed reinforcement particle size of 0.75 mm, 200 realizations were generated by randomly varying volume fraction of reinforcement particles from 10% to 30% using the procedure described in the previous sub-section.

Total 400 simulations (200 with hard reinforcement + 200 with soft reinforcement) were performed for SENB specimen with a set of 200 realizations. SENB specimen geometry with 3490 Q8 elements (2.6×10^4 DOFs). The material properties of matrix and reinforcement particles used for the analysis are tabulated in Table 1 and Table 2 respectively. To increase computational efficiency, the simulations were terminated after the detection of peak load for SENB specimen. For hard reinforcement particles, the correlation between volume fraction and peak load is positive as shown in left column of Fig. 12(b). With an increase in volume fraction of hard reinforcement particles from 10% to 30%, the peak load of SENB specimen increases by 12%. The peak load of SENB specimen dropped by 2% with an increase in volume fraction of soft reinforcement particles from 10% to 30%.

3.2. Mode-II problem: Single edge notched (SEN) specimen

In this section, the mode-II failure mechanism in single edge notched specimen with reinforcement particles is demonstrated using XFEM-strain gradient damage model. Fig. 13(a) shows a schematic of SEN specimen with geometric dimensions, spatial distribution of reinforcement particles and boundary condition. Fig. 13(b) shows a structured mesh generated with 21,042 Q8 elements. The horizontal displacement was applied on top surface of SEN specimen to mimic pure shear loading under plane strain condition. The properties of matrix and reinforcement particles used in the simulation are tabulated in Tables 1 and 2 respectively.

The effect of reinforcement particles on response curve for SEN specimen was demonstrated using three different cases i.e., without reinforcement, with hard reinforcement and with soft reinforcement as shown in Fig. 13(c). The peak load of SEN specimen with hard reinforcement particles i.e., 15.7 kN is 12% and 22% higher than the peak load of SEN specimen without reinforcement particles and with soft reinforcement particles respectively. The slope in the elastic region and final failure displacement for all three curves are almost same. The slope of load-displacement curve in post-failure (in-elastic) region is different for each case. For each case, the damage evolution and non-local equivalent strain contour plot at three different loading stages is shown in Fig. 14. The path of non-local equivalent strain and damage is not smooth due to presence of reinforcement particles.

3.3. Mix mode problem: L-Shaped specimen

In this section, the mix-mode failure mechanism in L-shaped specimen with reinforcement particles is demonstrated using XFEM-strain gradient damage model. Fig. 15(a) shows a schematic of an L-shaped specimen with geometric dimensions, the spatial

Mix-Mode Problem: L-Shaped specimen

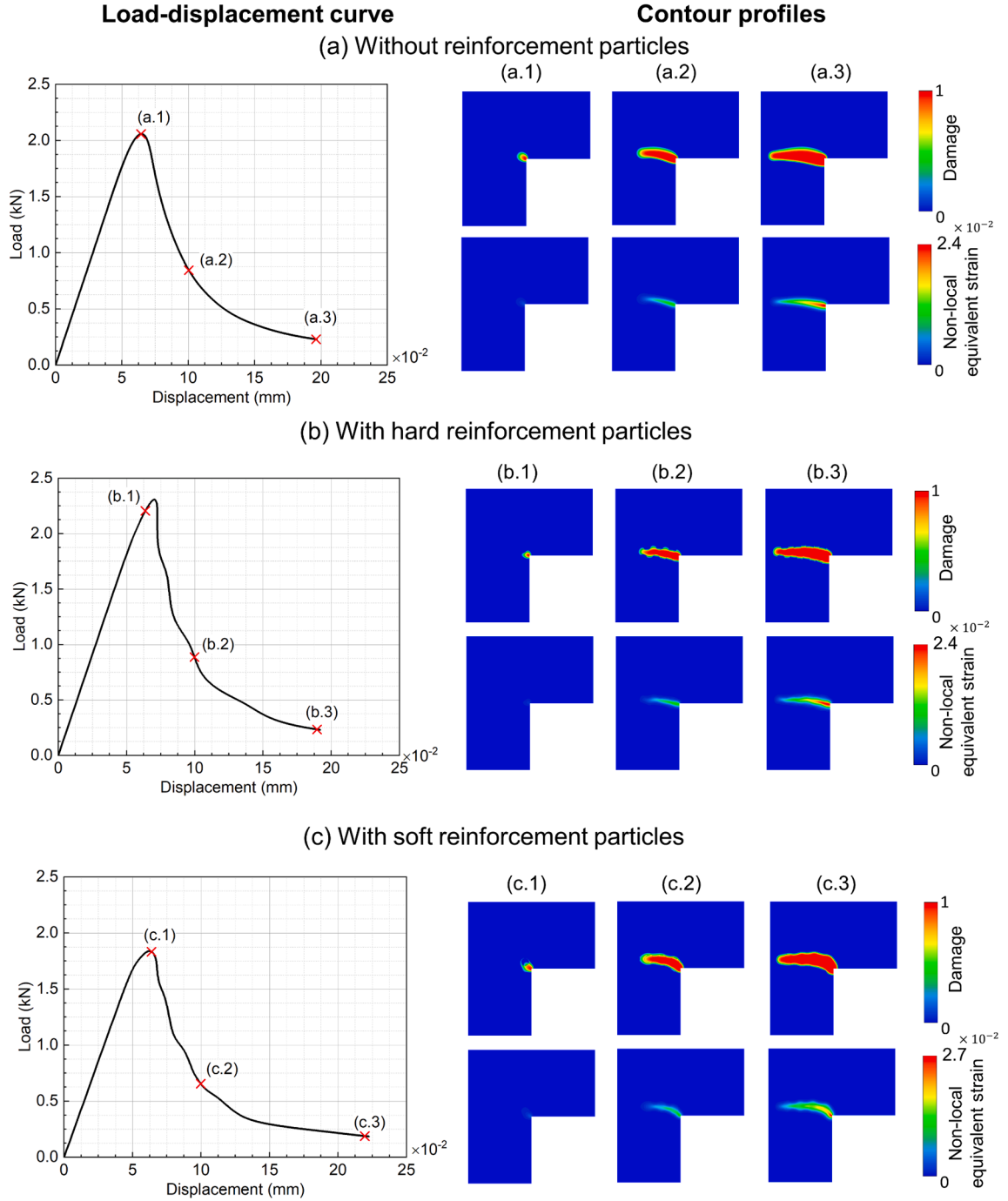


Fig. 16. The load–displacement curve along with contour plots (damage and non-local equivalent strain evolution) for L-shaped specimen with three different cases (a) without reinforcement particle (b) with hard reinforcement particle and (c) with soft reinforcement at three different loading stages.

distribution of reinforcement particles and boundary condition. Fig. 15(b) shows a structured mesh generated with 14,836 Q8 elements. The thickness of L-shaped specimen is 100 mm and simulated under plane strain condition. The properties of matrix and reinforcement particles used in the simulation are tabulated in Tables 1 and 2 respectively.

The effect of reinforcement particles on response curve for L-shaped specimen was demonstrated using three different cases i.e., without reinforcement, with hard reinforcement and with soft reinforcement as shown in Fig. 15(c). The peak load of L-shaped specimen with hard reinforcement particles i.e., 2.3 kN is 11% and 25% higher than the peak load of L-shaped specimen without reinforcement particles and with soft reinforcement particles respectively. The slope in the elastic region and final failure displacement for all three curves are almost same. The slope of load–displacement curve in post-failure (in-elastic) region is different for each case. For each case, the damage evolution and non-local equivalent strain contour plot at three different loading stages is shown in Fig. 16. The path of non-local equivalent strain and damage is non-smooth due to presence of reinforcement particles.

4. Conclusions

In the present study, coupling of XFEM and strain gradient damage model has been proposed to model damage initiation and growth in materials with reinforcement particles. The capability of XFEM-strain gradient damage model has been demonstrated by simulating Mode-I, Mode-II and mixed mode failure mechanism using SENB, SEN and L-shaped specimen with reinforcement particles. Elastic properties of matrix were considered from the experiment. Damage properties of material were calibrated by matching numerical response of SENB specimen with experimental response curve. The effect of soft and hard reinforcement particles on the response curve was evaluated for SENB, SEN and L-shaped specimen. Mesh-sensitivity analysis was performed to compare accuracy and computational performance of proposed XFEM model with the standard FEM model. Particle-sensitivity analysis was performed with the goal to quantify effect of size and volume fraction of reinforcement particles on the peak load of SENB specimen. The following conclusions are drawn from the analysis,

1. Due to presence of hard reinforcement particles, peak load for all specimen was increased at least by 11%.
2. The peak load for all specimen was decreased at least by 10% due to soft reinforcement particles.
3. Particle-sensitivity analysis predicts that the size of reinforcement particles (hard and soft) has a negligible (<3%) effect on the peak load of the SENB specimen.
4. The peak load of SENB specimen increases by 12% with an increase in volume fraction of hard reinforcement particles from 10% to 30%. The volume fraction of soft reinforcement particles has negligible (2%) effect on peak load of SENB specimen.
5. Mesh-sensitivity analysis shows that difference in peak load is negligible (less than 2%) between standard and proposed model. Both models have no significant difference (less than 3%) in area under response curve at higher mesh densities.
6. The computational performance analysis shows that the proposed model is at least 10% efficient (at higher mesh densities) than the FEM-strain gradient damage model for the SENB specimen with reinforcement particles.

The limitations of the present work are as follows,

1. The use of conventional strain gradient damage model in the proposed scheme leads to an unrealistic wide damage band in the simulations.
2. The gradient parameter (c^2) and the damage evolution parameter (β) in the proposed model significantly affect the results. Therefore, conclusions are limited to the problems presented in the study.
3. The proposed model is limited for modelling weak discontinuities.

In the future replacing the conventional strain gradient damage model used in the proposed model with the localizing gradient damage model [27] will resolve the issue of unrealistic wide damage band. Addition of variance-based sensitivity analysis [13] in the future study will quantify the influence of damage parameters on the solution and help us in outlining generalized conclusions. The experiments [41] shows that the presence of interphase layer influences the behavior of fiber reinforced ceramic matrix composites at high temperature. Therefore, an extensive analysis on effect of interphase thickness and properties on overall properties of composite with reinforcement particles is crucial. The analysis with interphase will be possible after above mentioned improvements in the proposed model.

Our long-term goal is to develop the framework to numerically predict variation in the properties of nuclear graphite. Nuclear graphite is highly heterogeneous quasi-brittle material (pores, reinforcement particles, micro-cracks) [21] and is used as a core component in high temperature Gen-IV reactor [19]. Nuclear graphite possesses large scatter in tensile, compressive, and flexural strength [14,7]. The computational viable 3D multiscale technique will be used for uncertainty quantification [39] of mechanical strength of nuclear graphite. Therefore, the proposed model acts as a groundwork for the future extension to 3D multiscale technique [28] with capability of modeling weak discontinuities at different scales including strong discontinuities [34] like micro-cracks. Experimental data from tensile, compressive, flexural and fracture toughness test of nuclear graphite will be used to validate the developed numerical model in the future.

CRedit authorship contribution statement

Manik Bansal: Visualization, Methodology, Formal analysis, Conceptualization, Writing – original draft, Writing – review &

editing. **Subrato Sarkar**: Writing – review & editing, Validation, Methodology, Data curation. **Indra Vir Singh**: Resources, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. von-Mises local equivalent strain

The expression of modified von-Mises local equivalent strain (in 2-D) can be written in terms of invariants as,

$$\varepsilon_{eq} = \frac{k_r - 1}{2k_r(1 - 2\nu)} I_1 + \frac{1}{2k_r} \sqrt{\frac{(k_r - 1)^2}{(1 - 2\nu)^2} I_1^2 + \frac{2k_r}{(1 - \nu)^2} J_2}, \quad (\text{A.1})$$

where

$$\begin{aligned} I_1 &= \varepsilon_{xx} + \varepsilon_{yy}, \\ J_2 &= 2 \left(\varepsilon_{xx}^2 + \varepsilon_{yy}^2 + 3\gamma_{xy}^2 - \varepsilon_{xx}\varepsilon_{yy} \right). \end{aligned} \quad (\text{A.2})$$

Through chain rule of differentiation, the derivative with respect to each tensorial strain would be,

$$\begin{aligned} \frac{\partial \varepsilon_{eq}}{\partial \varepsilon_{xx}} &= \frac{\partial \varepsilon_{eq}}{\partial I_1} \frac{\partial I_1}{\partial \varepsilon_{xx}} + \frac{\partial \varepsilon_{eq}}{\partial J_2} \frac{\partial J_2}{\partial \varepsilon_{xx}}, \\ \frac{\partial \varepsilon_{eq}}{\partial \varepsilon_{yy}} &= \frac{\partial \varepsilon_{eq}}{\partial I_1} \frac{\partial I_1}{\partial \varepsilon_{yy}} + \frac{\partial \varepsilon_{eq}}{\partial J_2} \frac{\partial J_2}{\partial \varepsilon_{yy}}, \\ \frac{\partial \varepsilon_{eq}}{\partial \gamma_{xy}} &= \frac{\partial \varepsilon_{eq}}{\partial I_1} \frac{\partial I_1}{\partial \gamma_{xy}} + \frac{\partial \varepsilon_{eq}}{\partial J_2} \frac{\partial J_2}{\partial \gamma_{xy}}. \end{aligned} \quad (\text{A.3})$$

The derivatives of equivalent strain with respect to invariants is written as,

$$\begin{aligned} \frac{\partial \varepsilon_{eq}}{\partial I_1} &= \frac{k_r - 1}{2k_r(1 - 2\nu)} + \frac{(k_r - 1)^2}{2k_r(1 - 2\nu)^2 \sqrt{\bar{A}}} I_1, \\ \frac{\partial \varepsilon_{eq}}{\partial J_2} &= \frac{2k_r}{(1 - \nu)^2 \sqrt{\bar{A}}}, \end{aligned} \quad (\text{A.4})$$

where $\bar{A} = \frac{(k_r - 1)^2}{(1 - 2\nu)^2} I_1^2 + \frac{2k_r}{(1 - \nu)^2} J_2$.

The derivatives of invariants with respect to tensorial strains is computed from Eq. (A.2) as,

$$\begin{aligned} \frac{\partial I_1}{\partial \varepsilon_{xx}} &= \frac{\partial I_1}{\partial \varepsilon_{yy}} = 1, \quad \frac{\partial I_1}{\partial \gamma_{xy}} = 0, \\ \frac{\partial J_2}{\partial \varepsilon_{xx}} &= 4\varepsilon_{xx} - 2\varepsilon_{yy}, \quad \frac{\partial J_2}{\partial \varepsilon_{yy}} = 4\varepsilon_{yy} - 2\varepsilon_{xx}, \quad \frac{\partial J_2}{\partial \gamma_{xy}} = 12\gamma_{xy}. \end{aligned} \quad (\text{A.5})$$

Appendix B. Non-local equivalent strain

The continuum damage model implemented herein is based on the non-local theory, whose underlying idea is to take spatial average of the local equivalent strain (ε_{eq}) over a local averaging domain area (V),

$$\bar{\varepsilon}_{eq}(\bar{\mathbf{x}}) = \frac{\int_V W(\zeta) \cdot \varepsilon_{eq}(\bar{\mathbf{x}} + \zeta) dV}{\int_V W(\zeta) dV}, \quad (\text{B.1})$$

where $W(\zeta)$ is defined as

$$W(\zeta) = \frac{1}{\sqrt{2\pi} l} \cdot e^{-\frac{\|\zeta\|^2}{2l^2}}. \quad (\text{B.2})$$

The Gaussian weight function $W(\zeta)$ defined in Eq. (B.2) consists of averaging length parameter l and relative position vector ζ as illustrated in Fig. B.1(a). It may also be noted from Fig. B.1(b) that within the non-local domain defined for spatial averaging of equivalent strain using Gaussian weight function $W(\zeta)$ as defined in Eq. (B.2).

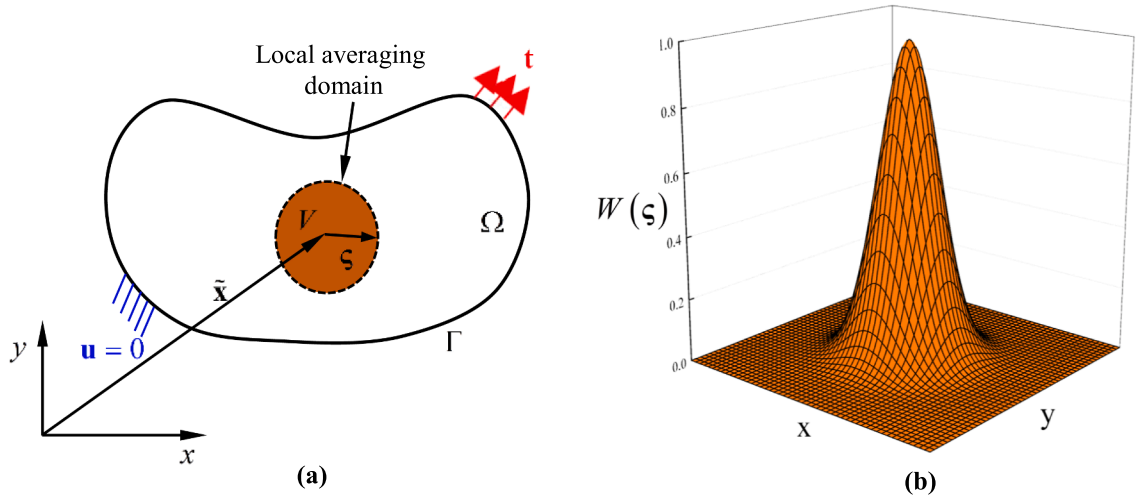


Fig. B1. (a) A schematic illustrating a local domain inside the problem domain used for averaging equivalent strain. The non-local equivalent strain is obtained after averaging equivalent strain quantities inside the local domain (b) surface plot of Gaussian weight function.

Appendix C. Quadratic and linear shape functions

In this section, details of quadratic and linear shape functions are presented. The quadratic shape functions and linear shape functions are used to approximate displacement and non-local equivalent strain respectively.

The shape functions of eight node quadratic quadrilateral element (Q8) are expressed as,

$$N_1 = \frac{1}{4}(\xi - 1)(1 - \eta)(1 + \xi + \eta), \quad (\text{C.1})$$

$$N_2 = \frac{1}{4}(1 + \xi)(\eta - 1)(1 - \xi + \eta), \quad (\text{C.2})$$

$$N_3 = \frac{1}{4}(1 + \xi)(1 + \eta)(\xi + \eta - 1), \quad (\text{C.3})$$

$$N_4 = \frac{1}{4}(\xi - 1)(1 + \eta)(1 + \xi - \eta), \quad (\text{C.4})$$

$$N_5 = \frac{1}{2}(1 - \xi)(1 + \xi)(1 - \eta), \quad (\text{C.5})$$

$$N_6 = \frac{1}{2}(1 + \xi)(1 + \eta)(1 - \eta), \quad (\text{C.6})$$

$$N_7 = \frac{1}{2}(1 - \xi)(1 + \eta)(1 + \xi), \quad (\text{C.7})$$

$$N_8 = \frac{1}{2}(1 - \xi)(1 + \eta)(1 - \eta). \quad (\text{C.8})$$

The shape function of four node linear quadrilateral element (Q4) are expressed as,

$$\bar{N}_1 = \frac{1}{2}(1 - \eta)(1 - \xi), \quad (\text{C.9})$$

$$\bar{N}_2 = \frac{1}{2}(1 - \eta)(1 + \xi), \quad (\text{C.10})$$

$$\bar{N}_3 = \frac{1}{2}(1 + \eta)(1 + \xi), \quad (\text{C.11})$$

$$\bar{N}_4 = \frac{1}{2}(1 + \eta)(1 - \xi). \quad (\text{C.12})$$

References

- [1] Agathos K, Ventura G, Chatzi E, Bordas SPA. Stable 3D XFEM/vector level sets for non-planar 3D crack propagation and comparison of enrichment schemes. *Int J Numer Meth Engng* 2018;113(2):252–76.
- [2] Bažant ZP, Gettu R, Kazemi MT. Identification of nonlinear fracture properties from size effect tests and structural analysis based on geometry-dependent R-curves 1991;28:43–51.
- [3] Bansal M, Singh IV, Mishra BK, Bordas SPA. A parallel and efficient multi-split XFEM for 3-D analysis of heterogeneous materials. *Comput Methods Appl Mech Engng* 2019;347:365–401.
- [4] Bansal M, Singh IV, Patil RU, Claus S, Bordas SPA. A simple and robust computational homogenization approach for heterogeneous particulate composites. *Comput Methods Appl Mech Engng* 2019;349:45–90.
- [5] Bansal M, Singh IV, Mishra BK, Sharma K, Khan IA. A stochastic XFEM model for the tensile strength prediction of heterogeneous graphite based on microstructural observations. *J Nucl Mater* 2017;487:143–57.
- [6] Bansal M, Singh IV, Mishra BK, Sharma K, Khan IA. A two-scale stochastic framework for predicting failure strength probability of heterogeneous materials. *Compos Struct* 2017;179:294–325.
- [7] Bansal M, Singh IV, Mishra BK, Sharma K, Khan IA. A numerical prediction of flexural strength probability for NBG-18 nuclear grade graphite using strength pair model. *J Strain Anal Eng Design* 2017;52(3):204–11.
- [8] Comi C, Mariani S, Perego U. An extended FE strategy for transition from continuum damage to mode I cohesive crack propagation. *Int J Numer Anal Meth Geomech* 2007;31(2):213–38.
- [9] Cazes F, Coret M, Combescure A, Gravouil A. A thermodynamic method for the construction of a cohesive law from a nonlocal damage model. *Int J Solids Struct* 2009;46(6):1476–90.
- [10] Cazes F, Simatos A, Coret M, Combescure A. A cohesive zone model which is energetically equivalent to a gradient-enhanced coupled damage-plasticity model. *Euro J Mech-A/Solids* 2010;29(6):976–89.
- [11] Geers MGD, de Borst R, Brekelmans WAM, Peerlings RHJ. Strain-based transient-gradient damage model for failure analyses. *Comput Methods Appl Mech Engng* 1998;160(1–2):133–53.
- [12] Geelen RJM, Liu Y, Dolbow JE, Rodríguez-Ferran A. An optimization-based phase-field method for continuous-discontinuous crack propagation. *Int J Numer Meth Engng* 2018;116(1):1–20.
- [13] Hamdia KM, Silani M, Zhuang X, He P, Rabczuk T. Stochastic analysis of the fracture toughness of polymeric nanoparticle composites using polynomial chaos expansions. *Int J Fract* 2017;206(2):215–27.
- [14] Hindley MP, Mitchell MN, Blaine DC, Groenwold AA. Observations in the statistical analysis of NBG-18 nuclear graphite strength tests. *J Nucl Mater* 2012;420(1–3):110–5.
- [15] Jirásek M. Non-local damage mechanics with application to concrete. *Rev Français Génie Civil* 2004;8(5–6):683–707.
- [16] Jirásek M. Mathematical analysis of strain localization. *Rev Europ Génie Civil* 2007;11(7–8):977–91.
- [17] Jin W, Arson C. XFEM to couple nonlocal micromechanics damage with discrete mode I cohesive fracture. *Comput Methods Appl Mech Engng* 2019;357:112617. <https://doi.org/10.1016/j.cma.2019.112617>.
- [18] Liu Z, Oswald J, Belytschko T. XFEM modeling of ultrasonic wave propagation in polymer matrix particulate/fibrous composites. *Wave Motion* 2013;50(3):389–401.
- [19] Marsden BJ, Jones AN, Hall GN, Treffi M, Mummery PM. Graphite as a core material for Generation IV nuclear reactors. In: *Structural materials for generation IV nuclear reactors structural materials for generation IV nuclear reactors*. Elsevier; 2017. p. 495–532. <https://doi.org/10.1016/B978-0-08-100906-2.00014-8>.
- [20] Mazars J, Pijaudier-Cabot G. From damage to fracture mechanics and conversely: a combined approach. *Int J Solids Struct* 1996;33(20–22):3327–42.
- [21] Mitchell BC, Smart J, Fok SL, Marsden BJ. The mechanical testing of nuclear graphite. *J Nucl Mater* 2003;322(2–3):126–37.
- [22] Msek MA, Cuong NH, Zi G, Areias P, Zhuang X, Rabczuk T. Fracture properties prediction of clay/epoxy nanocomposites with interphase zones using a phase-field model. *Engng Fract Mech* 2018;188:287–99.
- [23] Nguyen VP, Stroeven MARTIJN, Sluys LJ. Multiscale continuous and discontinuous modeling of heterogeneous materials: a review on recent developments. *J Multiscale Model* 2011;03(04):229–70.
- [24] Orifici AC, Herszberg I, Thomson RS. Review of methodologies for composite material modelling incorporating failure. *Compos Struct* 2008;86(1–3):194–210.
- [25] Patzák B, Jirásek M. Process zone resolution by extended finite elements. *Engng Fract Mech* 2003;70(7–8):957–77.
- [26] Patil RU, Mishra BK, Singh IV. A multiscale framework based on phase-field method and XFEM to simulate fracture in highly heterogeneous materials. *Theor Appl Fract Mech* 2019;100:390–415.
- [27] Poh LH, Sun G. Localizing gradient damage model with decreasing interactions. *Int J Numer Meth Engng* 2017;110(6):503–22.
- [28] Putar F, Sorić J, Lesičar T, Tonković Z. A multiscale method for damage analysis of quasi-brittle heterogeneous materials. *Comput Model Eng Sci* 2019;120(1):123–56.
- [29] Rabczuk T, Belytschko T. Cracking particles: a simplified meshfree method for arbitrary evolving cracks. *Int J Numer Meth Engng* 2004;61(13):2316–43.
- [30] Rabczuk T, Belytschko T. A three-dimensional large deformation meshfree method for arbitrary evolving cracks. *Comput Methods Appl Mech Engng* 2007;196(29–30):2777–99.
- [31] Rabczuk T, Zi G, Bordas S, Nguyen-Xuan H. A simple and robust three-dimensional cracking-particle method without enrichment. *Comput Methods Appl Mech Engng* 2010;199(37–40):2437–55.
- [32] Ren H, Zhuang X, Rabczuk T. Dual-horizon peridynamics: a stable solution to varying horizons. *Comput Methods Appl Mech Engng* 2017;318:762–82.
- [33] Roth S-N, Léger P, Soulaïmani A. A combined XFEM-damage mechanics approach for concrete crack propagation. *Comput Methods Appl Mech Engng* 2015;283:923–55.
- [34] Sarkar S, Singh IV, Mishra BK. A simplified continuous-discontinuous approach to fracture based on decoupled localizing gradient damage method. *Comput Methods Appl Mech Engng* 2021;383:113893. <https://doi.org/10.1016/j.cma.2021.113893>.
- [35] Sukumar N, Chopp DL, Moës N, Belytschko T. Modeling holes and inclusions by level sets in the extended finite-element method. *Comput Methods Appl Mech Engng* 2001;190(46–47):6183–200.

- [36] Simone A, Wells GN, Sluys LJ. From continuous to discontinuous failure in a gradient-enhanced continuum damage model. *Comput Methods Appl Mech Engng* 2003;192(41–42):4581–607.
- [37] Seabra MRR, Šuštarčič P, Cesar de Sa JMA, Rodič T. Damage driven crack initiation and propagation in ductile metals using XFEM. *Comput Mech* 2013;52(1): 161–79.
- [38] Tamayo-Mas E, Feliu-Fabà J, Casado-Antolin M, Rodríguez-Ferran A. A continuous-discontinuous model for crack branching. *Int J Numer Meth Engng* 2019;120 (1):86–104.
- [39] Vu-Bac N, Rafiee R, Zhuang X, Lahmer T, Rabczuk T. Uncertainty quantification for multiscale modeling of polymer nanocomposites with correlated parameters. *Compos B Engng* 2015;68:446–64.
- [40] Wang Y, Waisman H. From diffuse damage to sharp cohesive cracks: a coupled XFEM framework for failure analysis of quasi-brittle materials. *Comput Methods Appl Mech Engng* 2016;299:57–89.
- [41] Wang Yi, Zhang Ao, Li G-d, Liu S-P, Xiang Y, Cheng H-F. Sintering temperature and interphase effects on mechanical properties of an oxide fiber-reinforced Al₂O₃-SiO₂ composite fabricated by Sol-Gel method. *Appl Compos Mater* 2021;28(2):321–39.
- [42] Peerlings RH, de Borst R, Brekelmans WM, de Vree J. Gradient enhanced damage for quasi-brittle materials. *Int J Numer Methods Eng* 1996;39(19):3391–403.
- [43] Moës N, Cloirec M, Cartraud P, Remacle JF. A computational approach to handle complex microstructure geometries. *Comput Methods Appl Mech Eng* 2003; 192(28–30):3163–77.